Article

A split attractor design for rapidly writing a navigational goal

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Recurrent attractor networks are widely thought to form the basis of working memory[1–3], but how they can be rapidly switched on and off is unclear[4–7]. Here we investigate stability and switching in a recurrent circuit of the fly navigation centre[8]. hΔK and PFG neurons are recurrently connected in a ring structure and exhibit shared persistent bump activity that turns on with odour and terminates at the end of a goal-directed run. Using whole-cell recordings, we show that persistence in hΔK depends on recurrence, and that hΔK receives slow recurrent excitation and fast inhibition from its synaptic partners. Computational modelling reveals that these synaptic dynamics yield persistent attractor dynamics over a range of synaptic strengths. Next we examine the mechanisms of rapid switching. We find that whereas both populations show similar activity during runs, they become decoupled during turns and rest. We can reproduce these differential dynamics in our model by using inhibition to dynamically uncouple activity in hΔK from PFG. When hΔK is inhibited, PFG neurons follow their inputs from the compass system; when hΔK is disinhibited, recurrent interactions lock this input into place, forming a heading memory. Consistent with this model, we find that inhibitory inputs onto hΔK increase during turns and are suppressed during odour and goal-directed runs. Our work reveals how disinhibition can serve as a gate to rapidly write an ongoing measurement to a recurrent circuit.

Distributed networks of recurrently connected neurons are widely thought to form the substrate for short-term working memory[1–3]. During working memory tasks, neurons in the prefrontal cortex and elsewhere exhibit persistent activity that turns on with cue presentation and terminates with behaviour[9,10]. The duration of persistence in these neurons is much longer than their intrinsic membrane time constants, arguing that persistence arises from network interactions[3]. Computational models that feature local recurrent excitation and global inhibition can recreate the types of persistent activity that are observed experimentally[1,11] (Fig. 1a). However, such networks exhibit a fundamental trade-off between stability and flexibility[1]. Networks configured to generate stable attractor dynamics are difficult to shut off once activated[5]. Computational models therefore often implement gates to control the timing of attractor dynamics[4,6,7,12]. How such gates might be implemented at a circuit and synaptic level is unclear.

The *Drosophila* brain offers a unique opportunity to study the function of recurrent circuits at a cellular level[13–15]. Connectomics describes the architecture of fly networks with synaptic and cellular precision[8,16,17], and precise genetic tools enable functional interrogation of the same networks[18]. The fan-shaped body (FB) is a highly recurrent part of the fly navigation centre called the central complex, and is thought to compute and encode allocentric goals[8,19–25]. The FB contains about 30 types of local neuron that interact through complex patterns of feedforward and recurrent connectivity[8]. Several FB local neuron types have been shown to encode goal direction signals for navigation as positionally stable bumps of activity across the columns of the FB[22–24,26–28]. FB output neurons compare these goal signals to an internal compass, computed in the ellipsoid body[29] to drive egocentric steering[19,24,25]. How goal signals in local neurons are dynamically regulated, and the function of local recurrent connections, are both unclear.

Here we investigate how stable activity and rapid switching are combined in a discrete recurrent microcircuit within the FB. When flies navigate towards food, brief encounters with odour drive persistent goal-directed runs that are biased in the upwind direction[26,30]. In a previous study[26], we showed that a group of FB local neurons called hΔK exhibit a stable bump of activity that turns on with odour and persists while the fly maintains a stable heading. Sparse activation of hΔK neurons caused flies to walk in repeatable directions relative to wind[22], whereas silencing led to less persistent trajectories after odour loss[26]. Together these data support the idea that bump activity in hΔK neurons represents a persistent goal signal for navigation.

hΔK neurons are embedded in a recurrent network reminiscent of classic models of working memory (Fig. 1a,b). First, hΔK neurons are tightly recurrently connected in a ring structure with a second group of FB local neurons called PFGs (Fig. 1b and Extended Data Fig. 1a–d,f). Second, PFG neurons receive direct structured inputs from compass neurons in the ellipsoid body, which represent heading as a bump of activity[13,29]. Each PFG neuron receives heading inputs tuned to a different segment of the world[8] (Fig. 1b and Extended Data Fig. 1d). Finally, hΔK and PFG neurons both receive global input from several types

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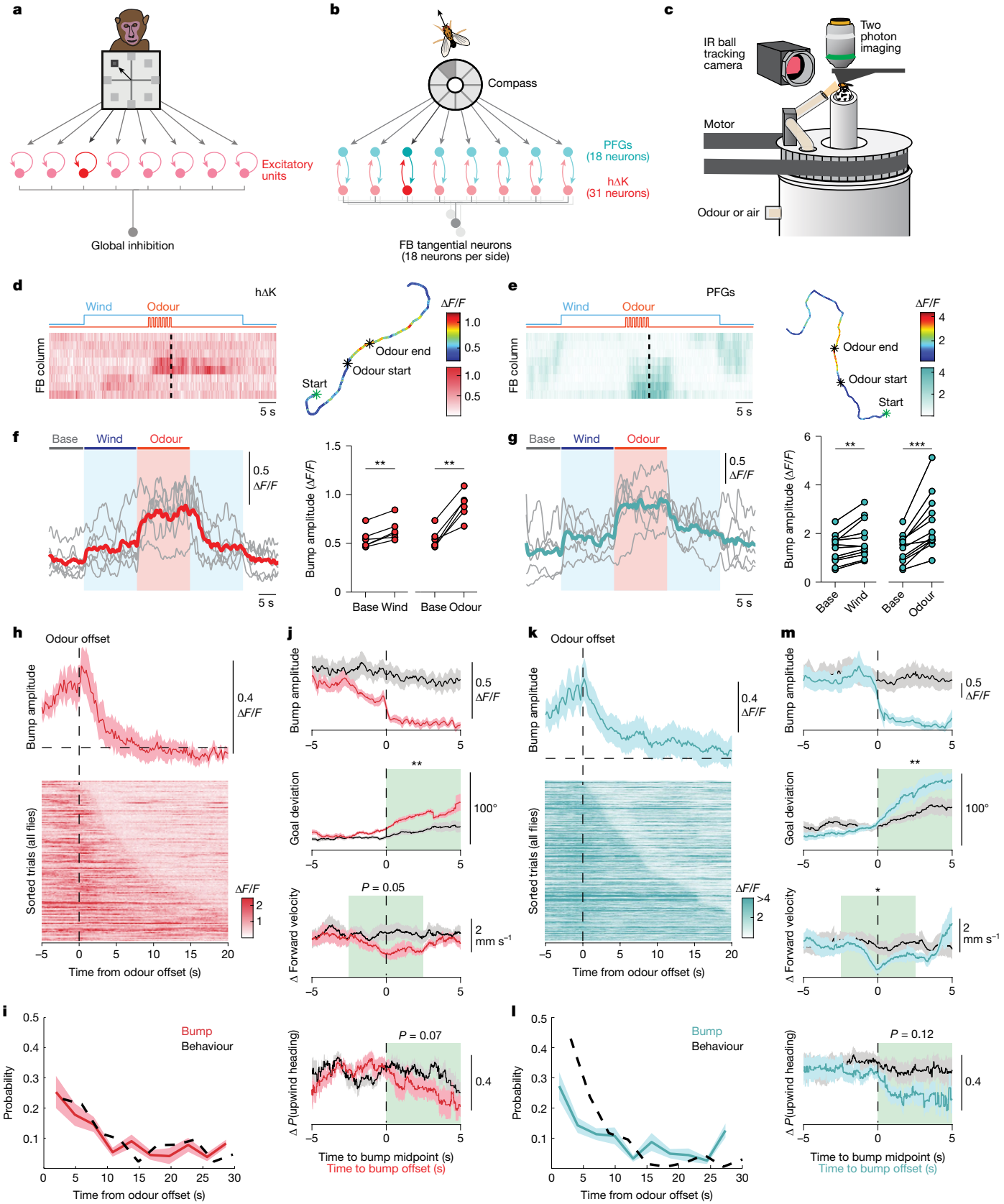


Fig. 1 | See next page for caption.

of broadly projecting tangential neurons (Fig. 1b and Extended Data Fig. 1e). The similarity of this network to classic working memory models suggests that this architecture might function to store heading representations and transform them into short-term goals.

Here we show through imaging, whole-cell physiology, genetic manipulations and computational modelling that persistent goal representations in hΔK are likely to emerge through recurrent interactions. We show that excitation and inhibition onto hΔK have distinct

Fig. 1 | Recurrently connected hΔK and PFG neurons show similar encoding of odour and goal-directed behaviour. a, Schematic of a recurrent network model developed to describe persistent spatial activity in macaque prefrontal cortex[1]. The recurrent network is composed of local recurrent excitation and global inhibition. **b**, Schematic of the recurrent network architecture between hΔK and PFG neurons in the FB. hΔK and PFG neurons display local recurrent excitation along the FB, PFG neurons receive spatially structured inputs from the compass, and both hΔK and PFG neurons receive global inputs from FB tangential neurons. **c**, Schematic of the in vivo imaging paradigm. Fly turning on an air-supported ball is yoked to the orientation of a tube that can deliver wind and odour (apple cider vinegar, ACV). Adapted from ref. 26, CC BY-NC-ND 4.0. **d**, Example of hΔK bump activity in response to 0.5% ACV. Left, activity shown as $\Delta F/F$ across FB columns as a function of time. Right, fictive 2D trajectory of the fly colour-coded by bump amplitude. **e**, Same as **d** but for PFG neurons. **f**, Mean bump amplitude of hΔK neurons during baseline, wind, and a 15 s pulse of ACV. Left, mean activity for each fly (grey) and across flies (red) as a function of time. Right, average activity for each fly as a function of stimulus phase ($n = 6$ flies). **g**, Same as **f** but for PFG neurons ($n = 12$ flies). Individual flies in grey and average in blue. **h**, Bump amplitude of hΔK neurons aligned to odour offset for all stimuli. Top, mean bump amplitude aligned to odour offset (mean ± s.e.m.). Bottom, individual trials sorted by duration of persistent activity. Red dashed line indicates odour offset and black dashed line indicates mean bump amplitude before odour onset. **i**, Distributions of hΔK bump persistence time (red; mean ± s.e.m.) and goal-directed walking (black) following odour offset. **j**, Mean bump amplitude, goal deviation, change in forward velocity and change in probability of upwind heading aligned to bump offset (red) or the centre of bump periods (grey) (mean ± s.e.m.). Green bars indicate the time bin for two-sided paired *t*-test. **k**–**m**, Same as **h**–**j** but for PFG neurons. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$; $P \geq 0.05$, not significant. Statistical details are in Supplementary Table 1.

dynamics that promote tuneable persistent attractor states across a range of excitation and inhibition weights. Finally, we examine the mechanisms that allow hΔK activity to be rapidly turned on by odour and off when the fly turns. We observe that even though hΔK and PFG are tightly anatomically coupled, they show similar activity only during runs, and diverge during turning and rest. We can reproduce these dynamics in a model by using feedforward inhibition to dynamically decouple hΔK from PFG in a stimulus and behaviour-dependent manner. Imaging from inhibitory inputs onto hΔK supports this model, showing that hΔK neurons are inhibited during turning and disinhibited during odour and goal-directed runs. Our data and models introduce a circuit motif—disinhibition of a split recurrent network—that allows stable attractor dynamics to be rapidly switched on and off by either sensory input or ongoing behaviour. We propose that this motif may represent a general solution to the problem of combining stability and flexibility in recurrent circuits.

Shared persistent activity in the hΔK–PFG network

hΔK and PFG neurons are strongly and reciprocally connected within columns of the FB (Fig. 1b and Extended Data Fig. 1f,g). Reciprocal connections between hΔK and PFG are the most numerous of any recurrent connection in the central complex (Extended Data Fig. 1h). hΔK neurons express the fast excitatory neurotransmitter acetylcholine, whereas PFGs express tyramine, myoinhibitory peptide and diuretic hormone 31, but no classical fast transmitter[18]. To test the hypothesis that these recurrent connections contribute to persistent bump activity previously observed in a broad hΔK line[26] (VT062617-Gal4), we imaged from each population using cell-type-specific split-Gal4 lines[18]. We recorded neural activity during closed-loop odour- and wind-guided navigation[26] (Fig. 1c). In this paradigm, a tethered fly walks on an air-supported ball that controls the orientation of a wind stimulus in closed loop to simulate wind from a fixed direction. During each 50 s wind trial, we presented odour (0.5% apple cider vinegar, ACV) as either a 15 s step, or as 1–10 500 ms pulses at 1 Hz. In response to odour, flies selected a heading trajectory that was biased upwind (Extended Data Fig. 1i–k) and persisted for several seconds after odour offset (Extended Data Fig. 1j,k).

Both hΔK and PFG neurons showed a localized bump of activity that was preferentially activated during odour (Fig. 1d–g and Extended Data Fig. 2a–c) and persisted while the fly maintained its trajectory (Fig. 1h–m and Extended Data Fig. 2d). In both populations, the probability and amplitude of bump activity grew similarly with pulse number (Extended Data Fig. 2a–c). In hΔK, the distribution of bump persistence times was very similar to the distribution of goal-directed walking times (Fig. 1i), whereas in PFG the neural distribution was slightly longer (Fig. 1l). To test whether neural persistence predicts behavioural persistence, we detected all bump offsets that occurred after the odour stimulus, and compared trajectories associated with bump offset to those associated with maintained bump activity (Fig. 1j,m). In both hΔK and PFG neurons, bump offset was associated with a significant deviation of heading from the 'goal' direction adopted during odour, compared with periods when the bump remained on. Bumps were also weakly associated with higher forward velocities, and a greater probability of orienting upwind (Fig. 1j,m). Together, our results argue that hΔK and PFG neurons show similar odour response dynamics and similar persistent activity associated with maintaining a goal heading, consistent with the hypothesis that recurrence between these populations contributes to persistent activity.

Synaptic contributions to persistent activity

We next investigated whether persistent activity might arise in part from intrinsic properties of hΔK neurons. We performed whole-cell patch clamp recordings from hΔK neurons (labelled by VT062617-lexA) and injected positive or negative current. We observed rapid responses to current injection (Fig. 2a and Extended Data Fig. 3a,b) with no persistent excitation. We conclude that intrinsic conductances alone do not generate persistent excitation in hΔK.

We next explored whether input from tangential neurons could drive persistence (see Fig. 2b, which shows connectivity and predicted neurotransmitters[8,31]). To test this, we recorded from hΔK neurons using VT062617-lexA (Fig. 2c) and stimulated tangential neurons that had been previously shown to encode odour and drive upwind running[22] (65C03-Gal4, labelling 17 tangential neurons including FB6A and FB6D (FB6A/D); Extended Data Fig. 3c). We observed a brief inhibition that switched to slow and persistent excitation later in the stimulus (Fig. 2c). Persistent excitation grew with duty cycle (Extended Data Fig. 3d,e), and was strongly reduced by the nicotinic antagonists curare (10 μM) and imidacloprid[32] (100 nM; Fig. 2d and Extended Data Fig. 3f). These tangential neurons express markers for both acetylcholine[18,22] and glutamate[18] (usually an inhibitory transmitter in the fly brain[33]) which might account for the biphasic effects of stimulation. To test whether recurrent signalling is required for persistent hΔK activity, we performed the same experiment while blocking recurrent signalling from hΔK using tetanus toxin (Fig. 2e). Blocking recurrent signalling abolished persistent excitation and switched the response of hΔK from excitation to weak inhibition (Fig. 2e,f). We conclude that hΔK neurons can generate persistent activity in response to synaptic input, that persistence requires cholinergic excitation, and that it depends on recurrent signalling by hΔK neurons.

To determine whether PFG neurons excite hΔK neurons, we used the same genetic strategy to stimulate PFGs and record from individual hΔK neurons. PFG stimulation drove excitation in some cells and inhibition in others (Fig. 2g), perhaps because optogenetic stimulation unrealistically depolarizes all members of the class. hΔK neurons that responded with excitation showed slow and persistent dynamics. Of note, we also observed slow persistent excitation in hΔK when we

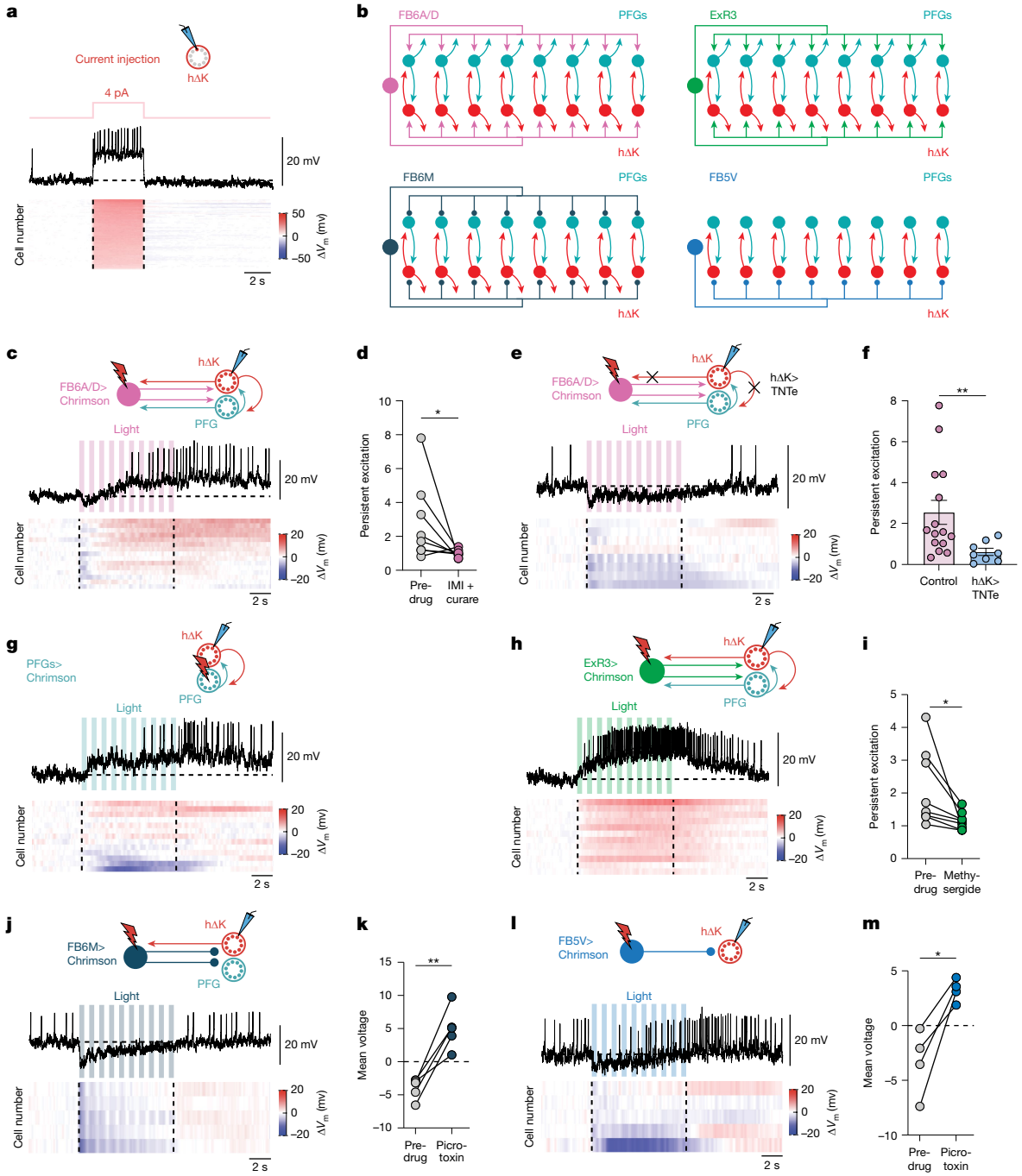


Fig. 2 | hΔK neurons receive slow excitation and fast inhibition from their tangential partners. a, Spiking and membrane potential responses of hΔK neurons (VT062617-lexA) during a 4 pA current injection. Top, example trace. Bottom, heat map of the mean membrane potential response with each cell as a row. **b**, Schematic of connectivity between tangential inputs (FB6A/D, ExR3, FB6M and FB5V) and hΔK and PFG neurons. **c**, Voltage response of hΔK neurons to optogenetic activation of FB6A/D (65C03-Gal4). Top, example trace. Bottom, heat map of the mean membrane potential response with each cell as a row. **d**, Persistent excitation (see Methods) following activation of FB6A/D before and after application of the nicotinic acetylcholine receptor blockers imidacloprid (IMI; 100 nM) and curare (10 μM) ($n = 8$ cells). **e**, Spiking and membrane potential response of hΔK neurons following optogenetic activation of FB6A/D when all hΔK neurons express tetanus toxin (TNTe) to block synaptic transmission. Top, example trace. Bottom, heat map of the mean membrane potential response with each cell as a row. **f**, Persistent excitation following activation of FB6A/D in control hΔK neurons (same data as **c**, $n = 15$ cells) and neurons expressing tetanus toxin ($n = 8$ cells). Data are mean ± s.e.m. **g**, Spiking and membrane responses of hΔK neurons following optogenetic activation of all PFG neurons (SS52590 split-gal4). Top, example trace. Bottom, heat map of the mean membrane potential response with each cell as a row. **h**, Spiking and membrane potential response of hΔK neurons following optogenetic activation of ExR3 (24C07-gal4). Top, example trace. Bottom, heat map of the mean membrane potential response with each cell as a row. **i**, Persistent excitation following activation of ExR3 before and after application of the general serotonin receptor antagonist methysergide (50 μM) ($n = 8$ cells). **j**, Spiking and membrane potential response of hΔK neurons following optogenetic activation of FB6M (49B08AD; VT063626DBD). Top, example trace. Bottom, heat map of the mean membrane potential response with each cell as a row. **k**, Mean voltage response of hΔK neurons during optogenetic stimulation of FB6M before and after adding picrotoxin (5 μM) ($n = 5$ cells). **l**, Spiking and membrane potential responses of hΔK neurons following optogenetic activation of FB5V (VT013602AD; 78H12DBD). Top, example trace. Bottom, heat map of the mean membrane potential response with each cell as a row. **m**, Mean voltage response of hΔK neurons during optogenetic stimulation of FB5V before and after adding picrotoxin (5 μM) ($n = 4$ cells; two-sided paired *t*-test). Statistical details are in Supplementary Table 1.

stimulated a serotonergic recurrent tangential neuron[34] (ExR3; Fig. 2b,h and Extended Data Fig. 3g,h). We could reduce this persistent excitation using the broad serotonin receptor antagonist methysergide (50 μM; Fig. 2h,i and Extended Data Fig. 3i). These data argue that multiple recurrent excitatory inputs onto hΔK drive slow and persistent dynamics.

In most computational models of recurrent attractor networks, broad inhibition is required together with local recurrent excitation to prevent runaway excitation[1,11,35]. We therefore tested whether some tangential inputs provide global inhibition onto hΔK. Using connectomic predictions of neurotransmitter phenotype[16,31], we identified two putative inhibitory tangential neuron types that synapse onto hΔK, called FB5V and FB6M (Fig. 2b), and made or identified lines that label each type (Extended Data Fig. 4a,b). To test whether these neurons provide functional inhibition onto hΔK, we stimulated each neuron type and recorded from hΔK using the same genetic strategy. In all lines we observed strong and rapid inhibition that depressed over time (Fig. 2j,l and Extended Data Fig. 4c). For both neuron types, stimulus-evoked inhibition was converted to excitation by 5 μM picrotoxin, a concentration that is known to block type A (GABA) γ-aminobutyric acid receptors[33] (Fig. 2k,m and Extended Data Fig. 4d). We conclude that some tangential inputs provide broad and rapid inhibition to the hΔK–PFG circuit, consistent with computational models of attractor network design.

Slow excitation stabilizes a persistent attractor

Imaging, physiology and connectomics indicate that the hΔK–PFG circuit contains the local recurrent excitation and broad inhibition required to form an attractor network. However, our experiments also revealed distinct dynamics for synaptic excitation and inhibition: excitation is slow, whereas inhibition is fast. Slow excitation might arise from peptides and modulators expressed by PFG and other FB neurons in addition to or instead of fast neurotransmitters[18]. To understand how synaptic dynamics contribute to recurrent network function, we developed a connectome-based computational model. In our model, the firing rates of individual hΔK and PFG neurons, and a single global inhibitory tangential neuron, depend on both external input and recurrent signalling between these neurons, as well as a leak term (Fig. 3a and Methods). The connections between the 30 hΔK and 18 PFG neurons were specified by a Gaussian function fit to the structure of recurrent connectivity between hΔK and PFG observed in the hemibrain connectome[8] (Fig. 3b and Extended Data Fig. 5a,b). The tangential neuron receives input from all hΔK and PFG neurons and provides global inhibition to both populations. The global strength of excitation and inhibition could be scaled through two free parameters (Fig. 3b).

We first explored whether this network could reproduce the persistent bump activity that we observed experimentally. We provided spatially structured input to hΔK and PFG neurons (Fig. 3c) and examined activity after this input was removed. We found that any asymmetric input to the network was transformed into a localized bump of activity with a profile resembling recorded bump activity (Fig. 3c,d). The width of the attractor bump grew and plateaued exponentially with the width of the Gaussian function describing local recurrent connectivity (Extended Data Fig. 5f,g). We also tested whether an attractor bump could be generated with real connectivity values. Using real weights from the hemibrain connectome[16], we could generate a bump at most locations in the FB, although bump amplitude and duration varied across locations (Extended Data Fig. 6a), whereas with Gaussian connectivity, we could generate a bump of equal magnitude and duration anywhere around the ring (Extended Data Fig. 6b). We conclude that the recurrent architecture of hΔK, PFG, and inhibitory tangential neurons supports persistent ring attractor dynamics even with noisy biological variation in connectivity strength[36].

Next, we investigated how differential synaptic dynamics contribute to network function. To model synaptic dynamics, we filtered the synaptic output of each population by a second-order differential equation to generate an alpha function (Methods). We fit the two time constants of these alpha functions to our synaptic physiology data (Extended Data Fig. 5c–e). When all synapses were fast, we observed that small changes in global excitation or inhibition strength rapidly transitioned the network from a regime with transient responses (decay <0.1 s) to a regime with non-decaying responses (Fig. 3e). Slow decays similar to those we observed experimentally were only observed for a narrow band of excitation and inhibition weights. By contrast, when recurrent excitation was slow, we observed a large regime of slowly decaying persistence (Fig. 3f). Within this regime, small changes in global excitation or inhibition strength gradually modulated the duration of persistence (Fig. 3g,h). By incorporating trial-by-trial noise in excitation and inhibition strengths, the slow model—but not the fast model—produced a distribution of decay times approaching the ones we observed experimentally (Fig. 3i). We conclude that the combination of slow excitation and fast inhibition allows the recurrent architecture of hΔK and PFG to produce persistent activity using a broad range of excitation and inhibition strengths, and further allows the duration of persistent activity to be tuned through small changes in these strengths.

Finally, we tested whether the effects of slow excitation on stability and tuneability are specific to the hΔK–PFG circuit architecture, or a more general property of recurrent circuits. We generated a simplified recurrent network with one self-recurrent excitatory unit and one bidirectionally connected inhibitory unit (Extended Data Fig. 7). By assuming that the membrane time constant and synaptic onset kinetics were much faster than synaptic offset kinetics, we created a dynamical system with only two variables: recurrent excitation and inhibition (Methods). Similar to our full model, this simple network showed persistent activity that depended on the speed of recurrent excitation: fast excitation led to a narrow band of excitation and inhibition strengths with decaying persistent activity (Extended Data Fig. 7a), whereas slow excitation led to a much larger regime of tuneable decay durations (Extended Data Fig. 7b). This regime deteriorates rapidly with excitatory synaptic speed but is relatively insensitive to the kinetics of inhibition (Extended Data Fig. 7c,d). We conclude that slow excitation promotes stable and tuneable persistence independent of network configuration.

Rapid uncoupling of the hΔK–PFG network

Our imaging, electrophysiology and modelling support the hypothesis that recurrent interactions between hΔK and PFG generate a shared persistent activity bump during odour-evoked goal-directed runs. However, this model does not address how the bump is turned on and off (Extended Data Fig. 8a). Close examination of our imaging data revealed that PFG and hΔK neurons showed distinct bump dynamics depending on the fly's behaviour. In PFGs, a low-amplitude bump was present at most times, and translated across columns during turns, becoming brighter during straight goal-directed runs (Fig. 4a). By contrast, hΔK only showed detectable bumps of activity during straight runs, and the bumps that appeared were generally fixed in place (Fig. 4b), consistent with our previous observations[26]. To test whether these differences in bump dynamics held across trials and flies, we measured how far each continuous bump deviated laterally across the FB (Fig. 4c). We found that both the maximum and cumulative lateral deviation were significantly smaller in hΔK compared to PFG (Fig. 4d). We further divided trajectories into periods of rest, turns, and straight running (Methods), and plotted the probability of observing a bump and its average amplitude during each behaviour. In both populations, bump probability and bump amplitude were highest during straight running (Fig. 4e,f). During turns and rest, PFGs showed a low-amplitude bump, whereas hΔK rarely showed a bump (Fig. 4e,f). These data argue that hΔK and PFG show similar dynamics during straight goal-directed runs, and that their activity must be functionally uncoupled at other times by some external mechanism.

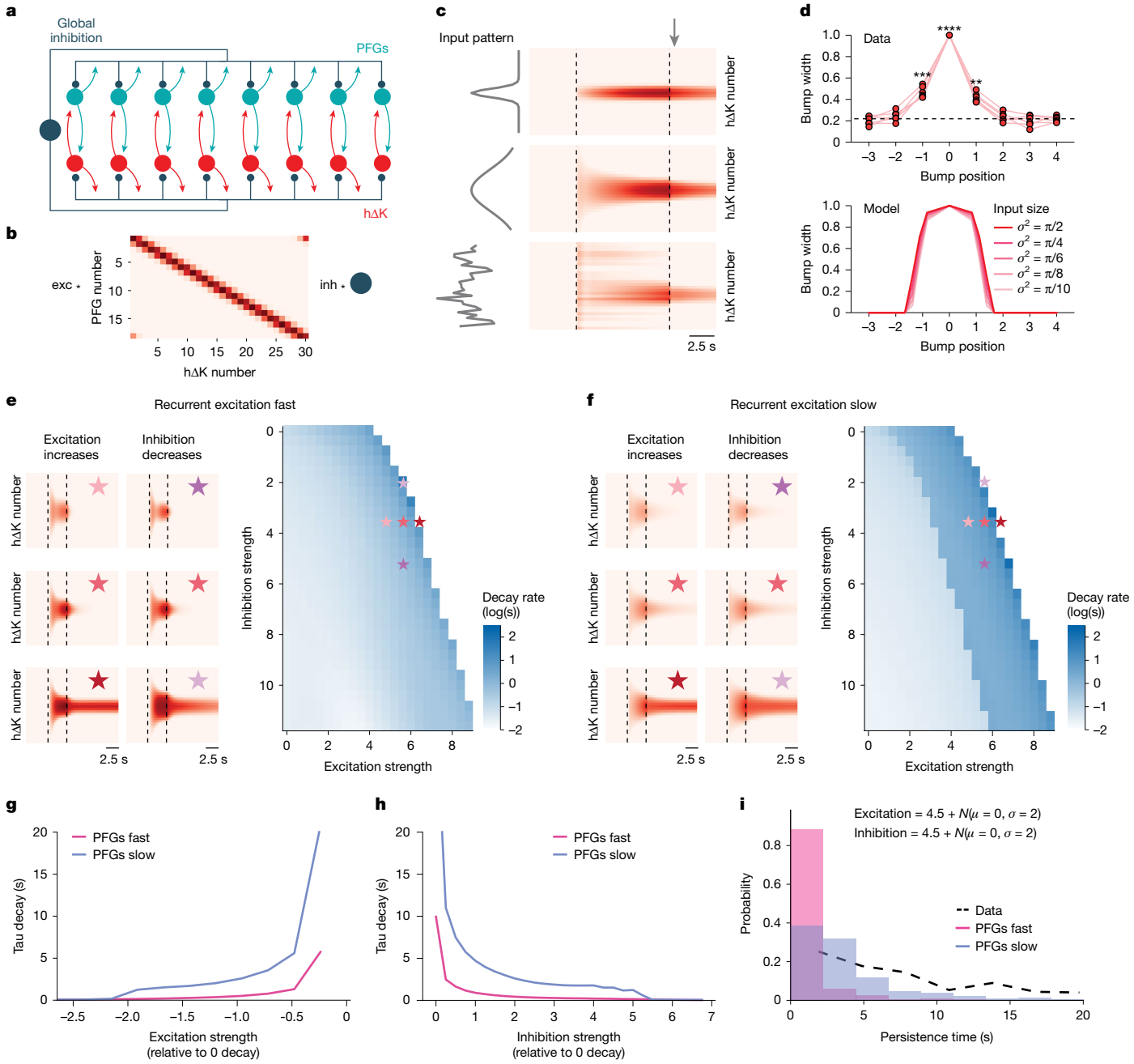


Fig. 3 | Slow recurrent excitation promotes more stable and tuneable attractor dynamics. a, Schematic of connectivity used in simulations. hΔK and PFG neurons are recurrently connected through ring recurrence and both populations are recurrently connected to a global inhibitory neuron. **b**, Schematic showing how the weight matrices are implemented in the model. The weight matrices between hΔK and PFG neurons are scaled by the parameter exc (excitation strength), whereas the connectivity of hΔK and PFG with the global inhibitory neuron are scaled by the parameter inh (inhibition strength). Asterisks refer to multiplication between scalars (exc or inh) and the synaptic weight matrix. **c**, Simulations of hΔK activity in response to narrow Gaussian input ($\sigma^2 = \pi/2$), wide Gaussian input ($\sigma^2 = \pi/10$) or random input. Black dashed lines indicate when the inputs are on. **d**, hΔK bump width from calcium imaging data (top) or estimated from model simulations (bottom; see Methods). **e**, Simulations of hΔK activity using fast recurrent excitation across various excitation and inhibition strengths. Right, decay rate ($\log_{10}$ scale) of the network following the cessation of input as a function of excitation and inhibition strength. White indicates zero network decay. Left, example dynamics at various locations in parameter space (stars). **f**, Same as **e** but using slow recurrent excitation in PFG neurons (see Methods). **g**, Mean decay time (tau) of the network as a function of excitation strength using fast (pink) or slow (blue) recurrent excitation. Averages are composed of rows from **e** (fast excitation) and **f** (slow excitation), each aligned to the smallest excitation strength that generated non-decaying persistence. **h**, Mean decay time (tau) of the network as a function of inhibition strength using fast (pink) or slow (blue) recurrent excitation. Averages are composed of the columns of **e** (fast excitation) and **f** (slow excitation), each aligned to the largest inhibition strength that generated non-decaying persistence. **i**, Distributions of persistence times in the fast (pink) and slow (blue) models when the excitation and inhibition strengths are randomly drawn from a Gaussian distribution. Simulations that generated non-decaying persistence were excluded. For comparison, the persistence distribution of hΔK activity following odour offset from Fig. 1i is shown in black. Statistical details are in Supplementary Table 1.

In many parts of the central complex, lateral movement of a bump has been shown to encode a change in heading[13,29], whereas in some local neurons, bump movement reflects travelling direction[20,21]. To test whether lateral shifts in PFG or hΔK bump position encode heading changes, we identified large shifts in heading (between 45–130°), and plotted bump position associated with these turns. We found that

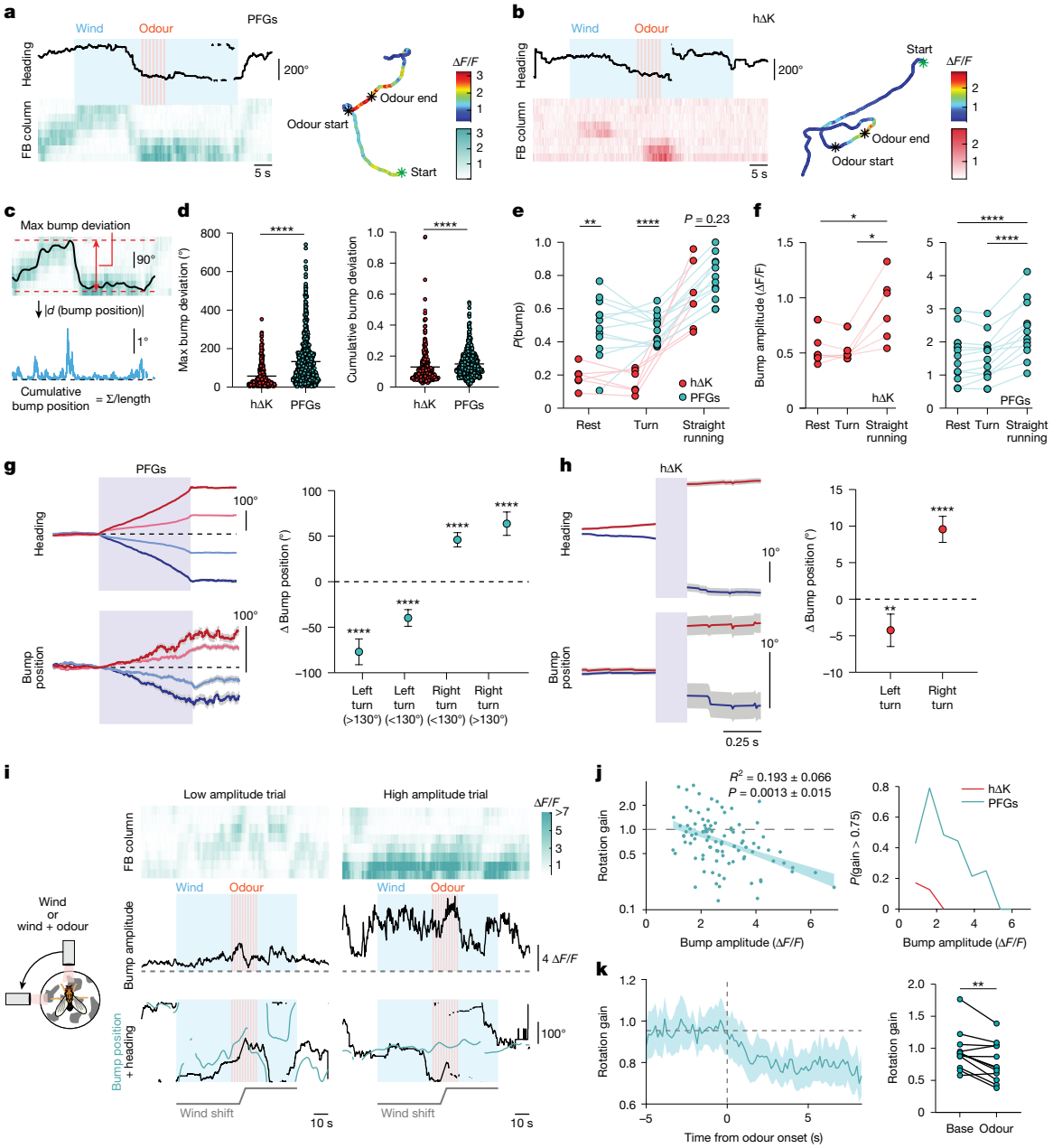


Fig. 4 | PFG and hΔK can show uncoupled dynamics depending on behavioural state. a, Example PFG bump activity during olfactory navigation. Left, heading and raw Δ*F*/*F* across FB columns as a function of time. Right, fictive 2D trajectory of the fly colour-coded by bump amplitude. **b**, Same as **a** but for hΔK neurons. **c**, Schematic showing how maximum (max) and cumulative bump deviation were computed (Methods). **d**, Maximum bump deviation (left) and cumulative bump deviation (right) for all hΔK bumps and PFG bumps. **e**, Bump probability for hΔK ($n = 6$ flies) and PFG ($n = 12$ flies) neurons across rests, turns and straight runs (Methods), compared using a two-way ANOVA with multiple comparisons. **f**, Bump amplitude for hΔK (left, $n = 6$ flies) and PFG (right, $n = 12$ flies) neurons across rests, turns and straight runs. **g**, Left, mean heading and PFG bump position during large right turns (dark red, Δheading > 130°, $n = 177$ turns), small right turns (light red, 130° > Δheading > 90°, $n = 198$ turns), small left turns (light blue, −90° > Δheading > −130°, $n = 111$ turns) and large left turns (dark blue, −130° > Δheading, $n = 139$ turns). Right, average shifts in PFG bump position across these turns (mean ± s.e.m.). **h**, Left, mean heading and hΔK bump position across subsequent bumps separated by right (red, 0° < Δheading, $n = 696$ turns) or left (blue, 0° > Δheading, $n = 476$ turns) turns. Right, average shifts in hΔK bump position across these turns (mean ± s.e.m.). **i**, Left, schematic for wind-shift experiment: wind or wind and odour were shifted by 90° over 2 s in a subset of trials. Right, example trials with wind shifts displaying low- (left) or high- (right) amplitude PFG bumps, each showing the raw PFG calcium data (top), bump amplitude (middle) and an overlay of PFG bump position and heading. **j**, Left, comparison of the rotation gain and average bump amplitude for PFG bumps present during the wind shifts with a linear fit computed across 10,000 bootstraps. Right, fraction of PFG and hΔK data with rotation gains over 0.75 across average bump amplitudes. For hΔK data, analysis was done on non-shift trials, but also see ref. 26. **k**, Rotation gain of PFG neurons over time aligned to odour onset (left) (mean ± s.e.m.) and compared between pre-odour and odour periods (right) ($n = 12$ flies). Statistical details are in Supplementary Table 1.

PFG bumps reliably shifted across columns during turns, similar to EPG neurons of the heading compass[29] (Fig. 4g). By contrast, hΔK activity largely disappeared during turns, but could reemerge at a new position following the turn (Fig. 4h). In single trails, we could observe a positive correlation of PFG bump position with heading (Extended Data Fig. 8b,d). By contrast, hΔK neurons showed discrete jumps in bump

position, with bumps appearing only during periods of relatively fixed heading (Extended Data Fig. 8c,d). Correlations with travel direction were weaker than those with heading for both populations (Extended Data Fig. 8e,f). We conclude that both PFG and hΔK can track heading changes, but that PFG encodes continuous changes in heading, whereas hΔK exhibits punctate encoding only during straight runs.

Our observations suggest that PFG neurons might be able to switch between tracking heading (at low amplitude) and holding a goal signal (at high amplitude), when hΔK neurons turn on as well. A goal signal should not shift with heading[23,24], and in a previous study, we found that hΔK bumps (labelled by the line VT062617-Gal4) did not shift when the wind was rotated during odour, driving compensatory turns[26]. To test whether PFG activity can switch between tracking heading and encoding a goal, we performed similar experiments in which we shifted the wind direction by 90° during either wind or odour while imaging from PFG neurons. We found that movement of the PFG bump depended on its amplitude (Fig. 4i,j). When PFG bump amplitude was low, bump position often shifted with heading (Fig. 4i, left), whereas when bump amplitude was high, bump position tended to remain fixed (Fig. 4i, right). To quantify this, we computed a rotational gain (ratio of bump movement to heading change) such that 1 indicates perfect encoding of heading and 0 indicates a positionally stable bump (see Methods). The rotational gain of the PFG bump was negatively correlated with bump amplitude on a trial-by-trial basis (Fig. 4j, left and Extended Data Fig. 8g). Rotational gains above 0.75 were only observed when the PFG bump had a low amplitude, and were almost never observed in hΔK (Fig. 4j, right). We also observed a significant drop in rotational gain following odour onset by using a sliding window to compute rotational gain over time (Fig. 4k and Methods). We conclude that whereas hΔK shows a fixed bump position when present, PFG can switch from following heading to maintaining a stable goal-like bump position, in a manner that depends on bump amplitude and sensory input.

Disinhibition can gate attractor dynamics

Our analysis suggest that the hΔK–PFG circuit can switch between a mode where PFG alone follows heading and a mode where both PFG and hΔK show stable goal-like bumps. To identify a mechanism underlying this switch, we revisited the connectome and our computational model. In published connectomes, only PFGs receive direct input from the compass system (EPG and Δ7 neurons that track fly heading, Fig. 5a; although hΔK does receive input from PEN neurons that encode angular velocity and rotate the heading bump[37,38]). This input is significantly weaker than the recurrent connections between hΔK and PFGs (Extended Data Fig. 9a,b). We investigated whether this difference in input could be combined with gating of recurrent connectivity to produce the dynamics that we observed in our imaging. If recurrent connections were shut off, PFGs should follow their compass input, whereas if recurrence were switched on, their activity would be locked in place at the location of the last compass input. To test this idea, we provided a moving compass input to PFG neurons, and gated recurrent connections between hΔK and PFG OFF or ON (Fig. 5b). Consistent with our prediction, we found that when recurrent excitation was gated OFF, PFG bump position followed the compass signal with low amplitude bumps, whereas hΔK was silent. By contrast, when recurrent excitation was gated ON, both PFG and hΔK showed a localized high-amplitude bump that remained in place despite shifting input, similar to our experimental observations (Fig. 4a,b,i–k).

Seeking a more physiological mechanism to rapidly gate connectivity between hΔK and PFG, we explored whether strong inhibition of hΔK could serve the same decoupling function. We repeated our simulation but replaced gating of recurrent connections with strong inhibition of hΔK (Fig. 5c). We found that when hΔK was inhibited, PFGs followed the compass with a low-amplitude bump, whereas when hΔK was disinhibited, both populations showed a localized high-amplitude bump. Consistent with the observation that recurrent connections between PFG and hΔK are several-fold stronger than feedforward input from the compass system across connectomes (Extended Data Fig. 9b), we found that in order for disinhibition to stabilize bump position, recurrent excitation strength must significantly outweigh feedforward heading input (Extended Data Fig. 9c,d).

Finally, we tested whether this disinhibitory mechanism could be used to rapidly write an ongoing heading signal into memory. We simulated time-varying heading trajectories and removed inhibition when the fly was oriented in different directions (Fig. 5d), or sequentially removed inhibition after an update in heading (Fig. 5e). We observed that disinhibition rapidly stabilized bump position at the location of the compass input when inhibition was first removed (Fig. 5f). We conclude that disinhibition could serve to functionally gate recurrent interactions between anatomically connected partners, rapidly switching the circuit from following an input to storing a pattern in memory.

Inhibitory inputs to hΔK show signatures of gating

Our model results suggest that disinhibition could support the divergent dynamics we observed in PFG and hΔK neurons. If this model is correct, we would expect inhibitory inputs onto hΔK to be tonically active, and to show inhibition during sensory stimuli that promote goal-directed running. We would also expect higher activity during turns, when we observe suppression of hΔK relative to PFG. To test these predictions, we imaged from the two inhibitory tangential neuron types we previously identified through electrophysiology– FB5V and FB6M– as flies performed our olfactory navigation task.

Activity in these inhibitory inputs to hΔK strongly supports our disinhibition model. We first imaged from FB5V neurons (Fig. 6a), a group of nine neurons per side that provides feedforward inhibition only onto to hΔK, similar to the inhibitory input that we modelled above (Fig. 5c). We observed tonic activity that was strongly supressed during odour, and continued its suppression during maintained trajectories after odour offset (Fig. 6b). Across flies, we observed weak inhibition by wind and strong inhibition by odour (Fig. 6c and Extended Data Fig. 10a). On a moment-to-moment basis, we found that FB5V activity was negatively correlated with upwind velocity, and positively correlated with angular speed, exactly as predicted by our disinhibition model (Fig. 6d,e). These encoding features might reflect the dendritic inputs to FB5V, which are found in regions encoding both odour (crepine (CRE)[39]) and pre-motor activity (lateral accessory lobe (LAL)[40–42]; Fig. 6a). To examine the encoding of turns in more detail, we detected large turns (Δheading > 130°) and plotted average activity associated with these (Fig. 6f). We observed strongly increasing activity during large heading changes (Fig. 6f,g). To examine encoding of goal-directed walking, we detected deviations from the goal direction adopted during odour (>90° from goal direction) and plotted activity associated with these deviations (Fig. 6h). We observed that FB5V activity increased when the fly deviated from its goal (Fig. 6h,i). Our imaging data thus support a role for FB5V suppression in enabling hΔK activity and goal-directed walking, and a role for FB5V elevation in decoupling hΔK from PFG during turns.

We also examined the activity of FB6M neurons. This group of two neurons per side provides input to both hΔK and PFG and receives feedback from hΔK (Fig. 6j). We observed tonic activity with suppression only in response to wind (Fig. 6k,l and Extended Data Fig. 10b). FB6M activity during wind was positively correlated with angular speed, but showed no correlation with upwind velocity (Fig. 6m,n). Similar to FB5V, FB6M showed increased activity during turns (Fig. 6o,p) and when the fly deviated from its goal direction (Fig. 6q,r). The wind encoding of FB6M might reflect its inputs in mechanosensory regions, which are known to encode the presence of wind (wedge (WED)[43,44]; Fig. 6j). In our model, we find that inhibition of both hΔK and PFG does not decouple their activity but instead shuts off both populations (Extended Data Fig. 9e). We conclude that FB6M also plays a disinhibitory role, but may

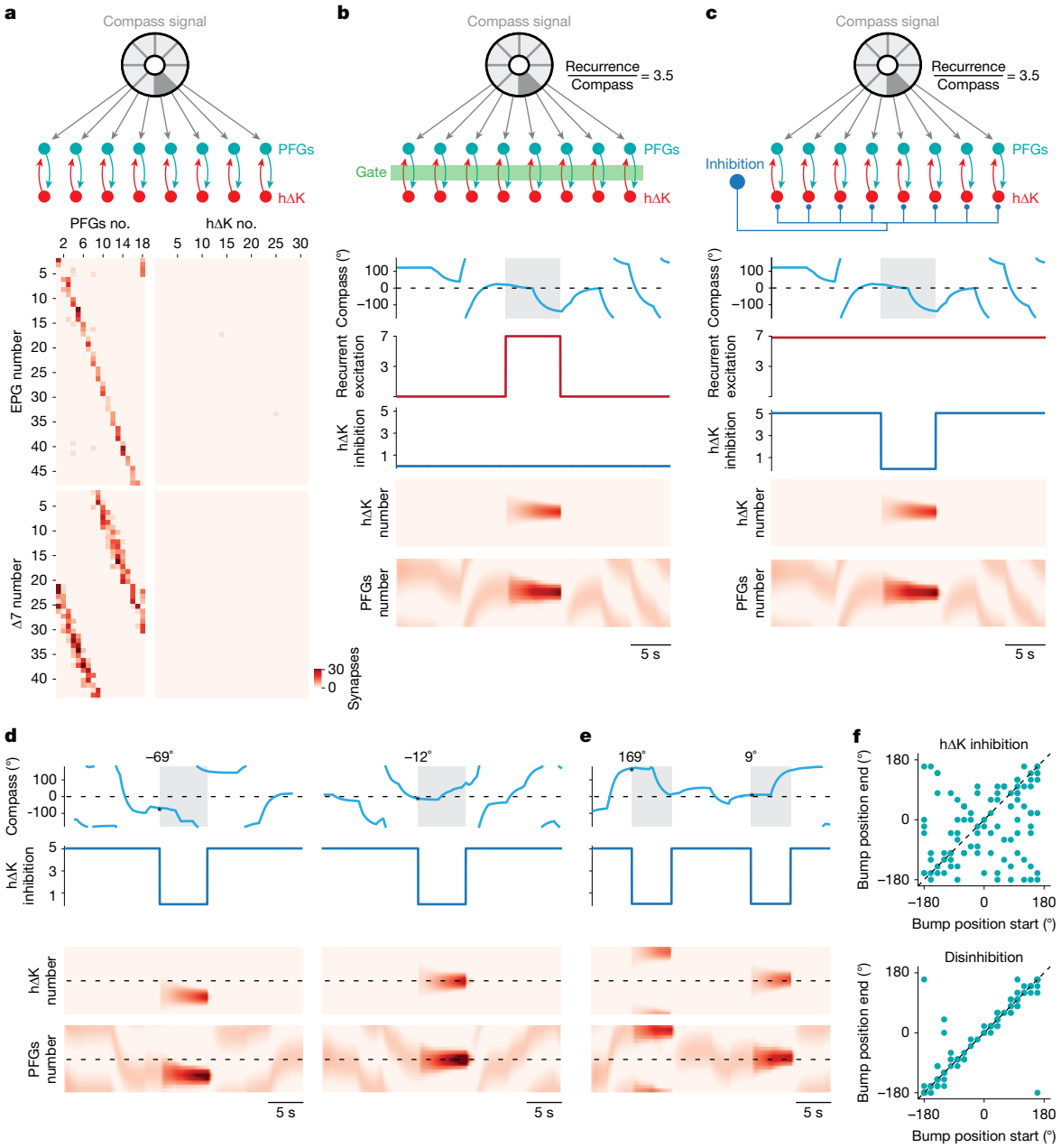


Fig. 5 | Disinhibition of hΔK enables rapid gating of recurrent attractor dynamics. a, Top, schematic of the connectivity between the compass system of the ellipsoid body and the hΔK–PFG recurrent network. Bottom, synaptic weights from heading-tuned EPG and Δ7 neurons to PFG (left) and hΔK (right) taken from the hemibrain connectome. hΔK and PFG neurons are sorted by their position along the FB, whereas EPGs are sorted by their position along the ellipsoid body and Δ7 neurons are sorted by their position along the protocerebral bridge. **b**, Simulations of hΔK and PFG population activity when the EPG compass signal (light blue) is randomly and stochastically updated and recurrent excitation strength (red) is transiently gated from OFF to ON (top). In these simulations, the strength of recurrent synapses relative to compass inputs is 3.5:1. Note the higher amplitude stable bump in the presence of recurrent excitation. **c**, Same as **b** but instead of directly modulating recurrent excitation, external global inhibition of hΔK neurons (dark blue) is transiently shut off. Note the higher amplitude stable bump during disinhibition. **d**, Further examples of the simulation shown in **c**, but with varying compass signal at the start of disinhibition (denoted by asterisk). Note that the stable hΔK and PFG bump positions depend on these initial compass signals (left, 120°; right, −12°). **e**, Same type of simulation as in **c**,**d** but with two bouts of disinhibition separated by a short time interval. Note that each disinhibition bout stabilizes the hΔK and PFG bumps to a different position that depend on the location of the compass input at the time of disinhibition (first, 169°; second, 9°). **f**, Bump position at the beginning and end of a period of hΔK inhibition (top) or disinhibition (bottom) as the compass input randomly fluctuates.

permissively regulate the activity of PFG and hΔK together, rather than serving to decouple these populations.

Discussion

Attractor networks composed of recurrently connected neurons contribute to many neural functions, including gaze stabilization[45,46], spatial working memory[9], motor control and decision making[10], navigation[13,47] and emotional states[48,49]. By forcing diverse patterns of input onto a limited set of stable states, attractor networks offer many advantages for computation[2], such as de-noising[50], evidence integration[51–53] and short-term information storage[1]. In computational models, recurrent networks with realistic fast units often fail to reproduce the slow persistent activity dynamics observed in experiments[35,48]. Attractor

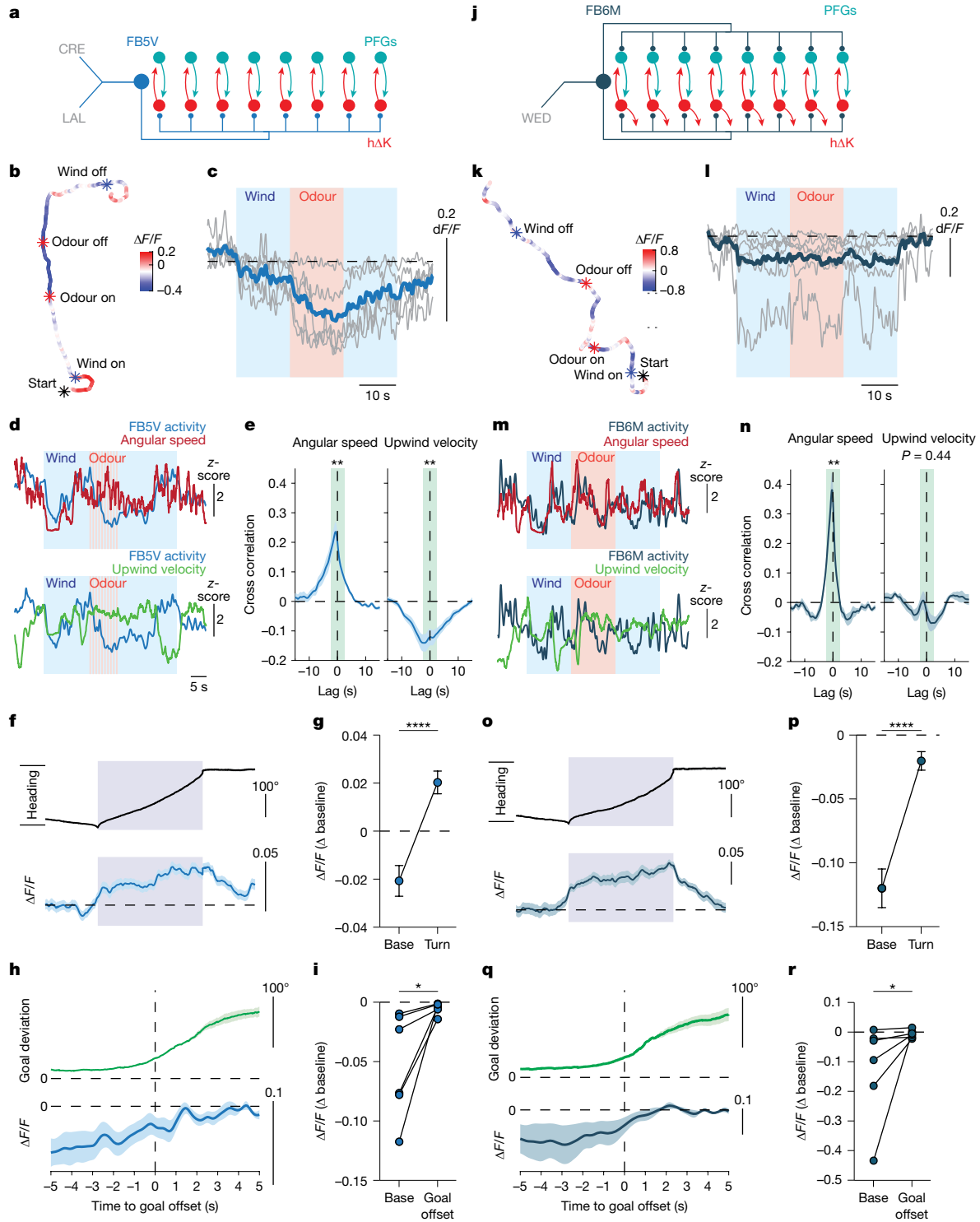


Fig. 6 | Inhibitory inputs to hΔK show signatures of gating. a, Schematic showing the connectivity between FB5V neurons and the hΔK–PFG recurrent network. Major brain regions innervated by dendrites of FB5V are shown in grey. **b**, Example walking trajectory colour-coded by activity in FB5V Δ*F*/*F* (baseline subtracted). Note the inhibition during odour that persists during continued upwind walking. **c**, Mean Δ*F*/*F* (baseline subtracted) of FB5V during baseline, wind, and 15 s of ACV. Individual flies are represented in grey, and the average across flies is in blue. **d**, Example of FB5V Δ*F*/*F*, angular speed and upwind velocity in response to wind and 10 pulses of ACV at 1 Hz. All data are *z*-scored. **e**, Cross-correlation between FB5V Δ*F*/*F* and angular speed (left) or upwind velocity (right). Statistics correspond to the average cross-correlation from −2 to 2 s, shown in green. **f**, Mean heading (absolute value and baseline subtracted) and FB5V Δ*F*/*F* during large (>130°) turns. Data are mean ± s.e.m. **g**, Mean FB5V Δ*F*/*F* before and at the end of the turns isolated in **f** (mean ± s.e.m., $n = 322$ turns). **h**, Deviation from goal direction and FB5V Δ*F*/*F* aligned to the offset of goal-directed walking following odour offset (Methods; mean ± s.e.m.). **i**, Mean FB5V Δ*F*/*F* during goal-directed walking (base) and following the offset of goal-directed walking ($n = 6$ flies). **j**–**r**, Same as **a**–**i**, but for FB6M ($n = 180$ turns in **p**; $n = 6$ flies in **r**). Statistical details are in Supplementary Table 1.

networks with fixed weights also do not generate activity restricted to particular behaviours, sensory inputs or internal states[10,49]. How biological recurrent networks are structured to generate stable dynamics that can be rapidly switched on and off by sensory input or behaviour is unclear.

One solution to generate robust slow dynamics is to make recurrent excitation slow[35,48]. A common mechanism for generating slow excitation is through *N*-methyl-D-aspartate (NMDA) receptors[35], although other mechanisms, such as synaptic facilitation[51] and peptide signalling[48] have also been proposed. Conversely, many network models use gates to control the onset and offset of attractor dynamics[4,6,7]. Disinhibition is a common motif for gating or regulating the gain of neural activity. In cortical circuits, a canonical disinhibitory circuit from VIP interneurons to PV interneurons to pyramidal neurons has been shown to regulate neuronal gain[54], contrast sensitivity[55] and learning[56]. Disinhibition also contributes to skilled motor actions coordinated by the basal ganglia in mice[57] and HVC in songbirds[58]. Despite the prevalence of disinhibition, the interaction of disinhibitory gating mechanisms with recurrent circuits has not been investigated in detail.

Here we took advantage of fly connectomes[8,16] and genetics[18] to investigate the relationship between anatomical connectivity and dynamics. Our results largely support the idea that recurrent connections between hΔK and PFG give rise to persistent activity bumps. We observed similar dynamics in both populations during goal-directed runs. We found that hΔK neurons exhibit persistent activity in response to synaptic input that was abolished by blockade of recurrent signalling. We built a connectome-based computational model of this circuit and found that it could stabilize activity at any location. However, some of the persistence we observed may arise upstream of this circuit, as inhibition in FB5V inputs to hΔK also outlasted the odour. A recent study identified MBONs upstream of FB5V that modulate movement timing following odours and might contribute to dynamics that we observed in FB5V[59]. The architecture of this network bears a marked resemblance to classical models that are proposed to underlie working memory in the prefrontal cortex[1,60]. However, the accessibility of the fly circuit also reveals several design features that enable this circuit to combine stable representations with rapid gating.

First, slow excitation in the hΔK–PFG circuit appears to be achieved through multiple biophysical mechanisms. PFG neurons do not express any fast transmitter, and instead express only neuromodulators and peptides[18]. Consistent with this expression profile, when we observed excitation in hΔK on stimulating PFGs, this excitation was slow and persistent (Fig. 2g). Our model indicates that slow recurrent excitation from local PFG neurons can generate attractor dynamics over a range of excitation and inhibition strengths and can make the duration of persistent activity tuneable. Our work expands on the classical idea that NMDA receptors are critical for building a stable working memory circuit[1,35], and recent results arguing that neuropeptide receptors are required for a persistent aggressive state in mice[48,61], suggesting that any slow signalling mechanism can serve to stabilize persistent activity.

Second, the hΔK–PFG recurrent network is split into two parts. In most attractor models, recurrent excitation occurs within a uniform population of excitatory neurons. Here instead we find that the network is composed of two morphologically[8] and transcriptomically[18,62] distinct neuron types that receive distinct inputs. PFG neurons receive a heading direction signal from the compass system, whereas hΔK neurons receive inhibition from FB5V. In our model, this allows PFGs to encode the content of memory, while FB5V gating of hΔK controls the timing of memory formation. By separating input from timing control, our model enables rapid gating of recurrent activity. hΔK and PFG also receive common input, for example from travel direction-encoding hΔB neurons[20,21], which might allow the circuit to perform additional computations. The separation of the circuit into two halves may also facilitate proper wiring of the network. Future studies might investigate the molecular signals that allow hΔK and PFG to wire together in a ring structure, that establish the necessary ratio between recurrent and feedforward synapses, and that allow hΔK and PFG to wire to both distinct and common inputs.

Finally we show through modelling that inhibition of one half of such a split attractor network can allow a circuit to rapidly switch between two modes: information following and information storage. Our model functions similar to a latch in digital logic[63]. When one half of the circuit (hΔK) is strongly inhibited, the recurrent circuit becomes functionally decoupled, and the other half (PFG) can follow their inputs freely. Disinhibition can rapidly turn on the recurrent circuit and lock the current input value into place. This model is consistent with several pieces of experimental data shown here. It reproduces the observed dynamics of both hΔK and PFG neurons during ongoing navigation and wind shifts, as well as the activity of FB5V during odour and turns. Our model is also consistent with causal experiments shown in our previous studies. In Matheson et al.[22], we showed that sparse activation of hΔK neurons (labelled by VT062617-Gal4) could drive movement in a reproducible goal direction, and in Kathman et al.[26], we showed that optogenetic silencing of hΔK neurons reduced the persistence of odour-evoked runs. A rigorous test of our computational model will require additional causal experiments. For example, activation and silencing of FB5V should switch the circuit between information following and information storage, with predictable changes in the amplitude and rotational gain of both PFG and hΔK activity. Writing a bump into hΔK or PFG using laser activation[24] should also produce predictable changes in the activity of the other population and behaviour.

Gated recurrent networks have been theorized to initiate intrinsic dynamics, such as persistent activity, and to coordinate more complex forms of dynamics, such as the manipulation of information in working memory and motor preparation and control[3,6]. Similar types of gates are used extensively in machine learning[64], including as a solution to rapidly reset memory traces[65]. Here we describe a biological form of gating that allows a recurrent anatomical circuit to dynamically alter its function on fast behavioural timescales. If integrated into recurrent networks with different connectivity structures[2,66–69], disinhibition could be used to rapidly alter the function of other types of attractor networks, allowing for flexible use of line attractors, rotations, sequences or bistable attractors in a stimulus- and behaviour-dependent manner.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-11144-9.

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Methods

Fly stocks

Female flies were raised on standard cornmeal-agar medium at 25 °C on a 12-h light cycle. For optogenetics experiments, the food was supplemented with 50 µl of 35 mM all-*trans* retinal (Sigma, R2500, dissolved in ethanol) mixed with ~1 teaspoon of hydrated potato flakes. Experimental flies were aged to 1–3 days for electrophysiology experiments or 9–14 days for imaging experiments. A list of fly genotypes is shown in Supplementary Table 2 and a list of fly reagents is shown in Supplementary Table 3.

Two-photon calcium imaging

Calcium imaging was performed as described in our previous study[26]. In brief, we anaesthetized flies over ice and a chilled (4–10 °C) aluminium sarcophagus and then glued them onto custom 3D-printed holders. We minimized movement of the brain by stabilizing the proboscis and head, and glueing the eyes, head and anterior tip of the thorax to the holder. To access the FB for imaging, we removed a small piece of the surface cuticle, trachea and air sacs over the back of the brain using sharpened forceps. We positioned flies over an air-supported ball (9 mm diameter General plastics FR-7110, painted with black spots, supported by 0.4 l s^{-1} air) using two cameras (30 fps, FLIR Blackfly, Computar 8.5 mm 1:1.3 adjustable lens) at the front and side of the fly. A third camera (100 fps, Grasshopper, 94 mm/0.5× InfiniStix Proximity Series lens, Edmund Optics Focal Length Extenders, working distance 15 cm) was used to track the ball using Fictrac software[70]. We illuminated the fly and ball using two infrared LEDs (M850F2) and T-cube LED Drivers (LEDD1B) coupled to fibre optic cables (Thorlabs M118L03 flat cleave patch cables). We continuously perfused (1 ml min^{-1}) the brain with extracellular saline (103 mM NaCl, 3 mM KCl, 5 mM TES, 8 mM trehalose dihydrate, 10 mM glucose, 26 mM $NaHCO_3$, 1 mM $NaH_2PO_4H_2O$, 1.5 mM $CaCl_2 2H_2O$ and 4 mM $MgCl_2.6H_2O$, pH 7.1–7.4, osmolarity 270–274 mOsm) bubbled with carbogen (5% CO_2, 95% O_2) and warmed to 33 °C.

We performed all imaging under a 20× water objective (Olympus Plan Fluorite XLUMP 20×/1.0) using an infrared laser (SpectraPhysics MaiTai HP or Coherent Axon 920) at 920 nM and 25–35 mW (measured at sample) using a galvo-galvo scanner. Simultaneous fluorescence emission from GCaMP7f and tdTomato were separated by bandpass filters (Semrock, FF01-525/50-32 for green and FF01-607/70-32 for red) and detected using two GaAsP photomultiplier tubes. We captured volumes consisting of three 192 × 96 µm optical sections, separated by 15 µm, taken at 8 volumes s^{-1} with a 1.2 µs dwell time.

In all imaging experiments, flies walked on a floating ball and received wind (25 cm s^{-1}) and odour (0.5% apple cider vinegar, 0.3 l min^{-1}) stimuli controlled by custom Python code. The wind direction was set by a rotary union (Dynamic Sealing Technologies, LT-2141-OF-ES12-F1-F2-C2) controlled by a stepper motor (Oriental PKP566FMN24A, with controller CVD524-K) and microcontroller (TeensyDuino +USB), and was controlled in closed loop by the flies' heading measured by ball movement as we described previously[26]. The starting wind direction was randomly selected from 0°, 45°, −45°, 135° or −135°. Wind was on from 5–60 s of each 65 s trial. The odour signal was randomly selected between five possible patterns on interleaved trials: odour step (15 s) or odour pulses of 1, 4, 7 or 10 pulses (1 Hz, 0.5 s pulse width). When the odour was turned off, a compensatory air valve was opened to maintain total air speed. Air to all lines (wind, ball support, odour, compensated air) were passed through a pressure regulator (Cole-Parmer MIR2NA) and charcoal filters (Drierite 26800 Drying column, with desiccant replaced by activated charcoal). Wind and compensated air were also humidified. Airflow for the wind line was controlled by a mass flow controller (Aalborg Model GFC17), while airflow for odour, compensated air, and ball support were controlled by flowmeters (Cole-Parmer PMK1-010608). We controlled the timing of wind stimuli using a solenoid valve (Cole-Parmer Masterflex 01540-11), and the timing of odour and compensate using high-speed three-way solenoid valves (Lee company, LHDA1233115HA).

During wind-shift experiments, the wind was rotated by 90° across 2.2 s either in the presence of odour (10 pulses at 1 Hz) or without odour. The onset of the shift was set to occur 4 s after odour onset. We also included control trials where wind was not shifted in the presence or absence of odour. The data from non-shift control trials were included in all analyses that did not depend on the precise odour stimulus.

Electrophysiology

For electrophysiology, we anaesthetized flies over ice, glued them to custom 3D-printed holders, and then removed their two front legs. To access dorsal cell bodies, we removed a large piece of the cuticle, trachea, and air sacs from the back of the brain, then manually detached the sheath using fine forceps. We minimized brain movements by removing the muscles on the brain and stabilizing the proboscis with glue. Starting at the dissection, and maintained throughout the entire experiment, we continuously perfused extracellular saline (same recipe as above), bubbled with carbogen, over the brain. Using a 40× objective (Olympus, LUMPLFLN40XW), an LED source (Cairn Research MONOLED), and a filter (U-N19002 AT-GFP/F LP C164404) we visualized GFP-positive cell bodies for recordings. Prior to recording, we cleaned the area around the target cell using fine-tip glass pipettes filled with extracellular saline.

For patch clamp recordings, we pulled glass pipettes with a Sutter P-1000 puller, pressure polished the pipettes (final resistance of 3–5 MΩ), and then filled them with intracellular solution (140 mM KOH, 140 mM aspartic acid, 10 mM HEPES, 1 mM EGTA, 1 mM KCl, 4 mM MgATP, 0.5 mM Na_3GTP, and 13 mM biocytin hydrazide). We amplified the voltage signals using an Axonpatch 200B amplifier with a Brownlee Precision 410 preamplifier and then digitized the signals at 10 kHz. We delivered 73 µW mm^{-2} red light (625 nm, measured at light source) for activating CsChrimson using a Thorlabs LED (M625F2) and T-Cube LED Driver (LEDD1B) coupled to optical fibres cables (Thorlabs M118L03 flat cleave patch cables) positioned below the head.

To better observe a mixture of excitation and inhibition, we depolarized cells to −38.1 ± 7.1 mV (mean ± s.d., $n = 37$ cells) using a small amount of positive current. Cells that did not spike were discarded. In all experiments, we randomly looped through 11 stimuli: seven optogenetic light stimuli (4 s of 12 µW mm^{-2} low power; 4 s of 23 µM mm^{-2} medium power; 4 s of 73 µM mm^{-2} high power; 1 Hz pulses with 0.2 duty cycle; 1 Hz pulses with 0.5 duty cycle; 1 Hz pulses at 0.8 duty cycle; and a light plume stimulus), three current stimuli (4 s of −4 pA; 4 s of 0 pA; and 4 s of 4 pA), and one wind or odour stimulus (wind on for 20 s, 5% apple cider vinegar on for 4 s). We constructed the light plume stimulus by filtering the odour encounters of a fly navigating in a virtual olfactory plume[26] with a positive derivative filter. The stimuli were looped five times in all cells, and then five times per drug condition. While we only report the results of light pulse stimuli in this paper, the voltage changes we observed—such as fast inhibition or slow persistent excitation—were seen across all optogenetic temporal patterns used.

In several experiments, we perfused drugs onto the brain to block neurotransmitter receptors. In these experiments, drugs were mixed with extracellular saline and perfused onto the brain for at least five minutes before resuming stimulus presentation. Since picrotoxin generated large excitatory responses to synaptic input, we often hyperpolarized the cells to −50.3 ± 10.2 (mean ± s.d., $n = 10$ cells) to minimize depolarization block. If cells did enter depolarization block, we paused the experiment until the cell recovered. The list of drugs, their sources and the final concentrations used are listed in Supplementary Table 4.

Immunohistochemistry

To perform immunohistochemistry, dissected brains were fixed in 4% paraformaldehyde (dissolved in PBS) for 15 min, washed 3× in PBS, incubated in 5% normal goat serum (dissolved in PBST) for 60 min,

incubated overnight in primary antibody solution (dissolved in 5% normal goat serum), washed 3× in PBST, incubated overnight in secondary antibody solution (dissolved in 5% normal goat serum), washed 3× in PBST, and washed 3× in PBS. The primary antibody solution contained chicken anti-GFP (Fisher Scientific RRID:AB_1074893) 1:50, mouse anti-nc82 (DSHB RRID:AB_2314866) 1:50, and rabbit anti-dsRed (Clontech 632496) 1:500. The secondary antibody solution contained Alexa488-conjugated goat anti-chicken (Fisher Scientific RRID:AB_2534096) 1:250, Alexa633-conjugated goat anti-mouse (Fisher Scientific RRID:AB_2535719) 1:250, and Alexa568-conjugated goat anti-rabbit (Fisher Scientific RRID:AB_2576217) 1:250. For imaging, the brains were mounted on microscope slides, immersed in Vectashield (Vector Labs H-1000), sealed with coverslips, and then imaged using a 20× objective (Zeiss W Plan-Apochromat 20×/1.0 DIC CG 0.17 M27 75 mm) on a Zeiss LSM 800 confocal microscope at 1.25 µM depth resolution.

To compare hemibrain neuron morphology to R65C03-GAL4 (Extended Data Fig. 2c), we warped a stain of R65C03 (from Flylight[71]) and hemibrain skeletons of layer 6 tangential neurons using the 2018 Janelia female template brain[72]. We overlaid the warped stain and warped skeletons and identified neurons with high overlap.

Data analysis

Connectomic analysis. Data from the hemibrain connectome[16] were obtained from neuprint explorer (http://neuprint.janelia.org, hemibrain:v1.2.1) and analysed using custom Python code. Prior to visualizing the weight matrices (Fig. 5a and Extended Data Fig. 1f), we sorted each cell type by their FB or protocerebral bridge x position (that is, columns), with hΔK positioning determined by their axons. To count the fraction of FB6M or FB5V synapses onto hΔK axons, we sorted synapses by their x position on a hΔK cell-by-cell basis. To calculate the ratio of recurrent synapse to EPG synapses (Extended Data Fig. 9b), we also examined data from the FAFB (https://codex.flywire.ai, FAFB v783) and male CNS (http://neuprint.janelia.org, male-cns:v0.9) connectomes. For each PFG neuron, we calculated the total number of synapses from hΔK neurons, the total number of synapses to hΔK neurons, and the total number of synapses from EPG neurons. The recurrence was the average synapses to and from hΔK neurons, and this was divided by the total inputs from EPGs to calculate the ratio. Since the FAFB data was less curated, and occasionally contained PFG neurons with 0 connectivity to EPG or hΔK neurons, we only included PFG cells that had a total recurrence strength and heading strength both above 10. Neuron and brain visualizations in Extended Data Fig. 1a,b were made using Flywire data[17,73] in Codex (https://codex.flywire.ai/?dataset=fafb&data_version=783)[74].

Calcium imaging and behaviour analysis. We used fictrac software[70] to transform the x, y and z rotations of a custom-made ball into 2D walking trajectories. The forward velocity and angular velocity were directly calculated as rotations around the x and z axes, respectively, which were then used to calculate heading (motor position, calculated as the integral of angular velocity), position (x = integral of forward velocity × sin(heading); y = integral of forward velocity × cos(heading)), and upwind velocity (v_y = forward velocity × cos(heading)). The rotation around the y axis (side velocity) was additionally used to calculate travel direction (heading + $\tan^{-1}$(forward velocity/side velocity)). For data analysis, the position, angular velocity, angular speed, upwind velocity, and forward velocity were smoothed using a running average of 1 s width.

We used custom Matlab software to pre-process our calcium imaging data (see ref. 26 for details), including motion correction, fluorescence extraction (across eight hand-drawn regions of interest (ROIs) across the FB), and alignment with fictrac behavioural data. To detect activity bumps in hΔK neurons, PFG neurons, and neurons labelled by VT062617-Gal4, we lowpass filtered bump amplitude data (Butterworth filter (sampling rate (fs) = 100 Hz, cut-off frquency (fc) = 2 Hz)) and then established a threshold on a trial-by-trial basis according to the minimum $\Delta F/F$ value of that trial and the $\Delta F/F$ s.d. across all trials. For most analyses, this threshold was set at minimum($\Delta F/F$) + 1 × s.d., but to identify large PFG bump onsets and offsets for Fig. 1k–m, the threshold was increased to minimum($\Delta F/F$) + 1.75 × s.d. A minimum bump duration was set at 3 s, and any above-threshold activity within 3 s of bumps was merged. To calculate bump position, the raw $\Delta F/F$ values across all 8 ROIs were smoothed (running average, width ~2 s) and then fit with a von Mises distribution at each time step using the Matlab function fminsearch. To aid fitting, we used the previous bump position or the ROI with the maximum $\Delta F/F$ (if the previous time step did not have a bump) as the initial search values. These bump positions were reported as a phase angle, where −180° is the left side of the FB and 180° is the right side of the FB.

During calcium imaging of FB5V and FB6M neurons, the calcium signal showed an artefactual decay from trial start to trial end. This artefact was removed by fitting an exponential decay using the beginning (first 5 s) and end (last 5 s) of the $\Delta F/F$ signal, and then subtracting this fit from the $\Delta F/F$ signal. Therefore, all $\Delta F/F$ signals reported for FB5V and FB6M are changes from this baseline.

In Fig. 1i,l, and Extended Data Fig. 1k, we calculated the persistence of walking direction following odour. To calculate the persistence of walking direction, we first defined a goal direction as the mean heading direction adopted in the first second after odour offset (post odour) or after trial start (baseline control), and then identified the time when flies stopped walking (forward velocity under 0.5 mm s^{-1} for at least 1.5 s) or deviated from the goal direction by at least 45°. In Fig. 1h,i,k,l, we calculated the persistence of neural activity following odour. To calculate the neural persistence, we calculated the time between odour offset and bump offset. To better compare neural and behavioural persistence, we only considered trials where the bump activity turned on before odour offset and turned off after odour offset.

In Fig. 1j,m, we aligned bump amplitude and behavioural data to bump offset. We only included trials where the bump activity turned on before odour offset and turned off after odour offset, and we removed any data before bump onset. In control bump ON data, we took the same trials, but only included data between bump onset and bump offset, each aligned by the midpoint of the bump. For goal deviation, the 'goal' was defined as the mean heading during the first 1/6th (between 0.18 and 4.46 s) of the bump ON or bump offset data. The durations of bump ON and bump offset data were matched for each trial.

In Fig. 3d, we calculated the mean bump width of hΔK activity. To do this, we isolated periods with bumps, aligned the raw $\Delta F/F$ calcium signals (across 8 ROIs) to the column with the highest signal, and then normalized the activity by the maximum activity.

In Fig. 4d, we calculated the maximum bump deviation and cumulative bump deviation. To quantify maximum bump deviation, we calculated the difference between the minimum and maximum unwrapped bump position between each bump onset and offset. To quantify cumulative bump deviation, we calculated how much the unwrapped bump position changed at each time point between bump onset and offset, summed the absolute value of these changes, and then divided the sum by the total length of the bump period.

In Fig. 4e–g and Fig. 6f,g,o,p, we identified discrete periods of turning. To identify turns, we identified segments of data where the unwrapped heading shifted by at least 90°. We then identified the beginning and end of each turn by searching backwards or forwards in time, respectively, until the heading stabilized (the maximum change in heading over 0.25 s was smaller than the 80th percentile of heading s.d. across all bins of the same duration). Since this algorithm isolated periods of turning with different durations, when turns were aligned in time (Figs. 4g and 6f,o), the data were time warped such that all turns were equal in length. In cases where small turns were compared to large turns (Fig. 4g), turns were classified as small if the total change in heading was less than 130°, and large otherwise.

In Fig. 4e,f, we further classified behaviour as 'rest' or 'straight running'. Rests were defined as periods of low movement ($v < 0.5$ mm s^{-1}) for at least 3 s. Straight running was defined as movements with high walking speed ($v >$ 50th percentile of speed for given fly) and stable heading (Δheading < 90°). Short straight running bouts (under 3 s) within 1 s of larger bouts were merged, and large bouts (over 3 s) within 3 s of each other were merged. Any turns that had overlapping sections with straight running bouts were removed.

In Fig. 4h, we calculated discrete shifts in hΔK bump position following turns. To do this, we identified pairs of hΔK bumps separated in time. For each bump pair, we calculated the change in wrapped heading from the end of the first bump to the beginning of the second bump, and then classified the pairs as left turns (0° > Δheading) or right turns (0° < Δheading). Since these data were wrapped, we centred the wrapping around the bump position or heading taken at the end of the first bump.

In Fig. 4i–k, we selected trials where a bump was present for the entire duration of the wind-shift stimulus and the fly walked for at least 20% of the bump period. We then calculated the rotational gain across these bump periods by dividing the cumulative change in unwrapped bump position by the cumulative change in unwrapped heading. Therefore, each data point in Fig. 4i represents a single bump from a single wind-shift trial. As both bump position and heading are represented by angles (see above for bump position), a rotation gain of 0 indicates stable bumps and a rotation gain of 1 indicates perfect tracking. The average bump amplitudes were calculated across the same time periods as the rotational gains. We ensured that the bump position and heading data were pre-processed equivalently by filtering both data sets using the same filters (moving average with a width of 13 samples at 100 Hz) and downsampling the data to match the original imaging sampling rate (downsample from 100 Hz to 6.37 Hz). To test the robustness of the correlation between bump amplitude and rotation gain, we bootstrapped the data (with replacement) 10,000 times and calculated s.d. error on the fit, R^2 and P values. For Fig. 4k, we analysed all trials that contained an odour stimulus. For each time point, the rotation gain was calculated by comparing the cumulative change in bump position to the cumulative change in heading within a 2 s window beginning at that timepoint. To avoid timepoints where the bump was mostly absent or the fly heading was stationary, we excluded timepoints if the total window did not contain sufficient bump data (<50%), if the cumulative delta heading was too low (<0.4°), or if the rotational gain was too high (>5).

In Extended Data Fig. 8b–f, we correlated bump position with heading or travel direction for each trial. We only excluded trials if the trial contained no bump activity.

In Fig. 6e,n, we cross-correlated FB5V and FB6M $\Delta F/F$ signals with angular speed or upwind velocity. Prior to cross-correlating, we mean subtracted the fluorescent and behavioural signals. To remove wind-coupled activity fluctuations in FB6M data, we only used data from 5 s after wind onset and 5 s before wind offset.

In Fig. 6h,q, we aligned FB6M and FB5V activity to deviations from goals. We first defined a goal direction as the mean heading adopted in the first second after odour offset, and then aligned neural and behavioural data to the time when flies deviated from this goal by at least 45°.

Electrophysiology analysis. For analysis of electrophysiology data, the voltage traces were lowpass filtered using a Butterworth filter (fs = 10,000 Hz, fc = 5 Hz), downsampled by a factor of 10, and then baseline subtracted. We calculated persistent excitation (Fig. 2d,f,i and Extended Data Fig. 3e,h) by summing the baseline-subtracted voltage signal between light offset and trial end, and then normalizing by the length of this period. For visualizing voltage signals across trials (for example, Fig. 2c), the voltage signals were sorted by persistent excitation (Fig. 2a,c,e,g,h and Extended Data Fig. 3a) or inhibition during the stimulus (Fig. 2j,l and Extended Data Fig. 4c).

Statistics and reproducibility. All statistics were ran using GraphPad Prism 10 and reported in Supplementary Table 1. Non-parametric statistical tests were used if the data were not normally distributed (see Supplementary Table 1). An alpha value of 0.05 was used to determine significance, with *$0.01 < P < 0.05$, **$0.001 < P < 0.01$, ***$0.0001 < P < 0.001$, ****$P < 0.0001$ reported on figures. For imaging experiments, flies were only included for further analysis if they reliably oriented upwind on at least 50% of trials, which was identified by a convergence of heading values towards the upwind direction (0°) and an increase in mean upwind velocity around wind or odour onset. For electrophysiology experiments, cells were only included for analysis if they showed visible spiking activity, defined as rapid deflections above 10 mV. Other than these criteria, all data and replications were included in the manuscript. Sample sizes (number of flies or cells) were selected based on previous experiments. Stimuli were randomized within each experiment. Experimenter was not blind to genotype but all analyses were applied programatically to all datasets without regards to genotype.

Modelling

Full FB model. To model the FB recurrent network, we created a rate-based dynamical system model consisting of 30 hΔK neurons, 18 PFG neurons, and 1 global inhibitory neuron. We approximated all dynamical systems using Euler's method of integration with a time step of 0.001 s. To generate ring recurrence between hΔK and PFG neurons, we gave each neuron a relative position in the FB, indicated by an angle from 0 to 2π, and then connected the populations with local excitation that was maximum at overlapping positions and decayed with distance. This was done explicitly using a Gaussian equation fit to the real connectivity from the hemibrain connectome (Figs. 3 and 5) or using the connectivity directly from the connectome (Extended Data Fig. 6). The local spread of the Gaussian connectivity was set by the local connectivity width (σ, Gaussian s.d.), which establishes how quickly the connectivity decays as the distance (in radians) between hΔK and PFG neurons increases ($\sigma_{PFG>h\Delta K} = 0.25$, $\sigma_{h\Delta K>PFG} = 0.25$, in radians). The best fit Gaussian connectivity was found by searching for the local connectivity width (σ) that minimized the absolute error between the fit weight matrix and real weight matrix. Each column of the real weight matrix (the connectivity of each PFG neuron to all hΔK neurons) was normalized to ensure fits were based on the local connectivity width rather than variability in synaptic strength across the FB. Although real hΔK neurons have a 180° shift across the FB, for simplicity, we only show connectivity and activity simulations from the 'axons' of hΔK neurons. The weight matrices between hΔK or PFGs and the global inhibitory neuron were either one-to-all or all-to-one. To simulate the neural activity of each population, we used the following ordinary differential equations:

$$\tau_{h\Delta K} \times \frac{d\vec{u}_{h\Delta K}}{dt} = -\alpha_{h\Delta K} \times \vec{u}_{h\Delta K} + \text{exc} \times W_{PFG,h\Delta K}\vec{U}_{PFG} - W_{inh,h\Delta K}\vec{U}_{inh} + \vec{I}_{h\Delta K} \quad (1)$$

$$\tau_{PFG} \times \frac{d\vec{u}_{PFG}}{dt} = -\alpha_{PFG} \times \vec{u}_{PFG} + \text{exc} \times W_{h\Delta K,PFG}\vec{U}_{h\Delta K} - W_{inh,PFG}\vec{U}_{inh} + \vec{I}_{PFG} \quad (2)$$

$$\tau_{inh} \times \frac{d\vec{u}_{inh}}{dt} = -\alpha_{inh} \times \vec{u}_{inh} + \text{inh} \times (W_{h\Delta K,inh}\vec{U}_{h\Delta K} + W_{PFG,inh}\vec{U}_{PFG}) + \vec{I}_{inh} \quad (3)$$

Where τ_x is the time constant for population x, α_x is the leak of population x, $\vec{u}_x$ is the activity of population x, exc is the excitation strength, inh is the inhibition strength, $W_{x,y}$ is the weight matrix from population x to population y, $\vec{U}_x$ is the synaptic output of population x, and $\vec{I}_x$ is

the external input to population x. τ_x and α_x were set to 1 and 10, respectively, for all populations. $\vec{I}_x$ was used to set the input patterns to each population.

To simulate the synaptic output of each population, we used the following second-order ordinary differential equations to simulate alpha functions:

$$\vec{U_x}'' = -(a_x + b_x) \times \vec{U_x}' + a_x \times b_x \times (F_x(\vec{u}_x) - \vec{U_x}) \quad (4)$$

Where $1/a_x$ is the decay rate of the alpha function for population x, $1/b_x$ is the growth rate of the alpha function for population x, and F_x is the activation function of population x. The synaptic growth and decay values used for fast and slow synapses were fit to synaptic physiology data by searching for the a and b values that minimized the absolute error between the fit activity and synaptic physiology voltage data. The fast synapses were fit to hΔK voltage responses to FB6M (49B08AD; VT063626DBD) pulsed stimulation, and were set to $a_x = 4.63$ and $b_x = 25$. The slow synapses were fit to hΔK voltage responses to FB6A/D (65C03-Gal4) stimulation in the presence of curare (10 μM) and IMI (100 nM), and were set to $a_x = 0.64$ and $b_x = 25$. We used the following activation functions:

$$F_{\mathrm{h\Delta k/PFG}}(u_x) = \begin{cases} 0 \text{ if } u_x < 0 \\ u_x \text{ if } 0 < u_x < 1 \\ 1 \text{ if } u_x > 1 \end{cases} \quad (5)$$

$$F_{\mathrm{inh}}(u_x) = \begin{cases} 0 \text{ if } u_x < 0 \\ u_x \text{ if } 0 < u_x \end{cases} \quad (6)$$

During the simulations in Fig. 3, sinusoidal (direct offset = 1), Gaussian (normalized such that max = 1), or random input (Gaussian distribution with $\mu = 1$, $\sigma = 0.5$) were provided to the hΔK and PFG neurons for 4 or 10 s. During the simulations in Fig. 5 and Extended Data Figs. 8a and 9e, we only provided input to PFG neurons, which was sinusoidal with a phase that we randomly and stochastically updated over time. Phase shifts were stochastic, occurring with a 0.1% chance at each time step, and when they occurred, they were randomly drawn from a Gaussian distribution with $\mu = 0$ and $\sigma = 50°$ and then smoothed over time according to:

$$\tau_{\mathrm{smooth}} \times \frac{\mathrm{dheading}_{\mathrm{smooth}}}{\mathrm{d}t} = \mathrm{heading}_{\mathrm{raw}} - \mathrm{heading}_{\mathrm{smooth}} \quad (7)$$

Where τ_{smooth} equals 1. To account for the differences in synaptic strength between EPGs>PFG and hΔK<->PFG synapses, we decreased the amplitude of the sinusoidal input to 1/6 of the recurrent excitation strength. We controlled the gate in Fig. 5 by either externally changing the excitation strength (exc in equations (1,2), increased from 0 to 7) or changing external input to hΔK ($\vec{I}_{\mathrm{h\Delta K}}$ in equation (1), inhibition corresponded with a value of −5, and disinhibition corresponded with a value of 0). In Extended Data Fig. 9e, we additionally attempted to gate bump dynamics by adding external input to hΔK and PFG neurons ($\vec{I}_{\mathrm{h\Delta K}}$ and $\vec{I}_{\mathrm{PFG}}$ in equation (1), increased from −1 or −2 to 0).

To calculate the network decay rate in Fig. 3, we calculated the maximum difference in hΔK output at each time step following external input offset and then fit this difference with an exponential decay function.

In Fig. 3d, we expanded the model simulations to predict how each simulation would appear if the neurons were recorded using population imaging, such as calcium imaging. Calcium imaging captures activity throughout each active neurons' axons. Therefore, we calculated how much each model neuron should spread across the FB to synapse with its postsynaptic partners. This spread was then applied to the model simulation to get a better prediction of how far activity of the active neurons would spread across the FB. In Extended Data Fig. 5f,g, to keep the neuron size the same across simulations we used real connectome data to estimate the width of real hΔK neurons, and then used this spread to estimate the width of FB activity. We fit the relationship between local connectivity s.d. and FB activity percentage with an exponential equation.

In the parameter space in Extended Data Fig. 7c,d, for each pair of parameters ($a_{\mathrm{inhibition}}$ and $a_{\mathrm{excitation}}$), we first calculated the decay tau of the excitatory neuron across each excitatory strength value, removed all values above 30 s, and then we summed these values to get the total 'sum'.

In Fig. 5f and Extended Data Fig. 6, we calculated the bump position by finding the relative position (as an angle) of the neuron with the maximum activity.

Simple recurrent network model. To simplify the recurrent network to two dynamic variables, we began by reducing the hΔK and PFG populations to one global excitatory neuron. This led to the following 6 variable dynamical system:

$$\tau_{\mathrm{exc}} \times \frac{\mathrm{d}u_{\mathrm{exc}}}{\mathrm{d}t} = -u_{\mathrm{exc}} + W_{\mathrm{exc,exc}}U_{\mathrm{exc}} - W_{\mathrm{inh,exc}}U_{\mathrm{inh}} + I_{\mathrm{exc}} \quad (8)$$

$$\tau_{\mathrm{inh}} \times \frac{\mathrm{d}u_{\mathrm{inh}}}{\mathrm{d}t} = -u_{\mathrm{inh}} + W_{\mathrm{exc,inh}}U_{\mathrm{exc}} + I_{\mathrm{inh}} \quad (9)$$

$$U_{\mathrm{exc}}'' = -(a_{\mathrm{exc}} + b_{\mathrm{exc}}) \times U_{\mathrm{exc}}' + a_{\mathrm{exc}} \times b_{\mathrm{exc}} \times (F_{\mathrm{exc}}(u_{\mathrm{exc}}) - U_{\mathrm{exc}}) \quad (10)$$

$$U_{\mathrm{inh}}'' = -(a_{\mathrm{inh}} + b_{\mathrm{inh}}) \times U_{\mathrm{inh}}' + a_{\mathrm{inh}} \times b_{\mathrm{inh}} \times (F_{\mathrm{inh}}(u_{\mathrm{inh}}) - U_{\mathrm{inh}}) \quad (11)$$

We made two simplifying assumptions. First, we assumed that the synaptic rise and decay constants ($1/a_x$, $1/b_x$) were much slower than the membrane time constants (τ_x). This means that the variables u_{exc} and u_{inh} instantaneously follow the synaptic dynamics. Second, we assumed that the synaptic decay time constant (a_x) was much slower than the synaptic rise time constant (b_x). These assumptions lead to the following two-variable dynamical system:

$$U_{\mathrm{exc}}' = a_{\mathrm{exc}} \times (F_{\mathrm{exc}}(W_{\mathrm{exc,exc}}U_{\mathrm{exc}} - W_{\mathrm{inh,exc}}U_{\mathrm{inh}} + I_{\mathrm{exc}}) - U_{\mathrm{exc}}) \quad (12)$$

$$U_{\mathrm{inh}}' = a_{\mathrm{inh}} \times (F_{\mathrm{inh}}(W_{\mathrm{exc,inh}}U_{\mathrm{exc}} + I_{\mathrm{inh}}) - U_{\mathrm{inh}}) \quad (13)$$

We used the same activation functions as those used for the excitatory and inhibitory neurons in the full model (equations (5,6)). During simulations, we kept $W_{\mathrm{exc,inh}}$ constant at 1, whereas we varied $W_{\mathrm{exc,exc}}$ and $W_{\mathrm{inh,exc}}$ with excitation strength and inhibition strength, respectively. During input, the excitatory unit received an external input equal to 1, whereas the inhibitory unit received no input.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data are available at Zenodo at https://doi.org/10.5281/zenodo.19687299 (ref. 75). Source data are provided with this paper.

Code availability

All code is available at Github at https://github.com/nagellab/Lanzetal2025.

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Author contributions A.J.L. and K.I.N. designed the study. A.J.L. performed all electrophysiology and imaging experiments, analysed all data, and performed all computational modelling. N.D.K. provided assistance and training on the two-photon imaging protocol and contributed to data analysis. E.H. performed confocal imaging. B.E. provided advice and feedback on computational modelling and developed the minimal recurrent circuit model. A.J.L. and K.I.N. wrote the paper with feedback from all authors.

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Competing interests The authors declare no competing interests.

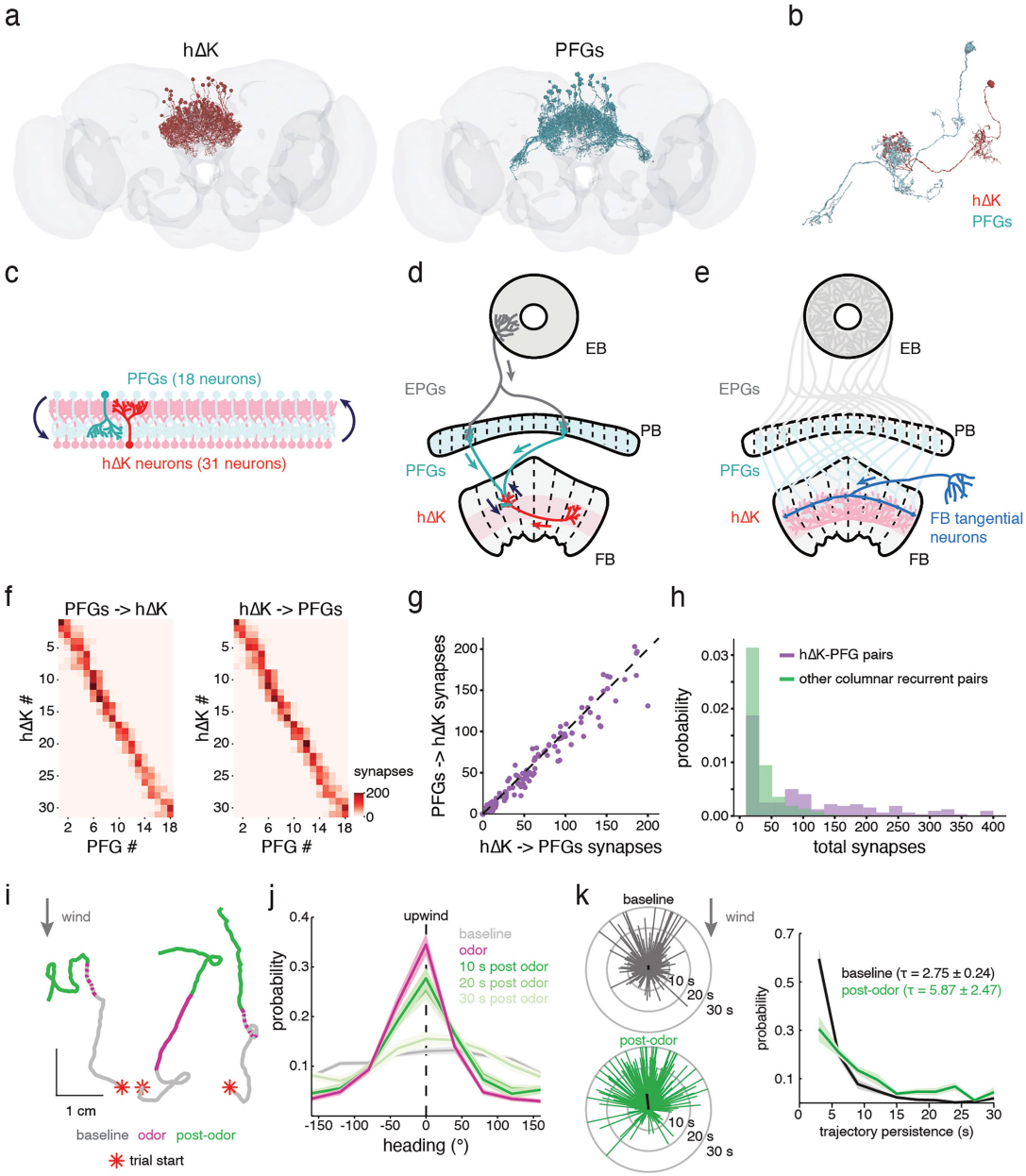
Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-026-11144-9.

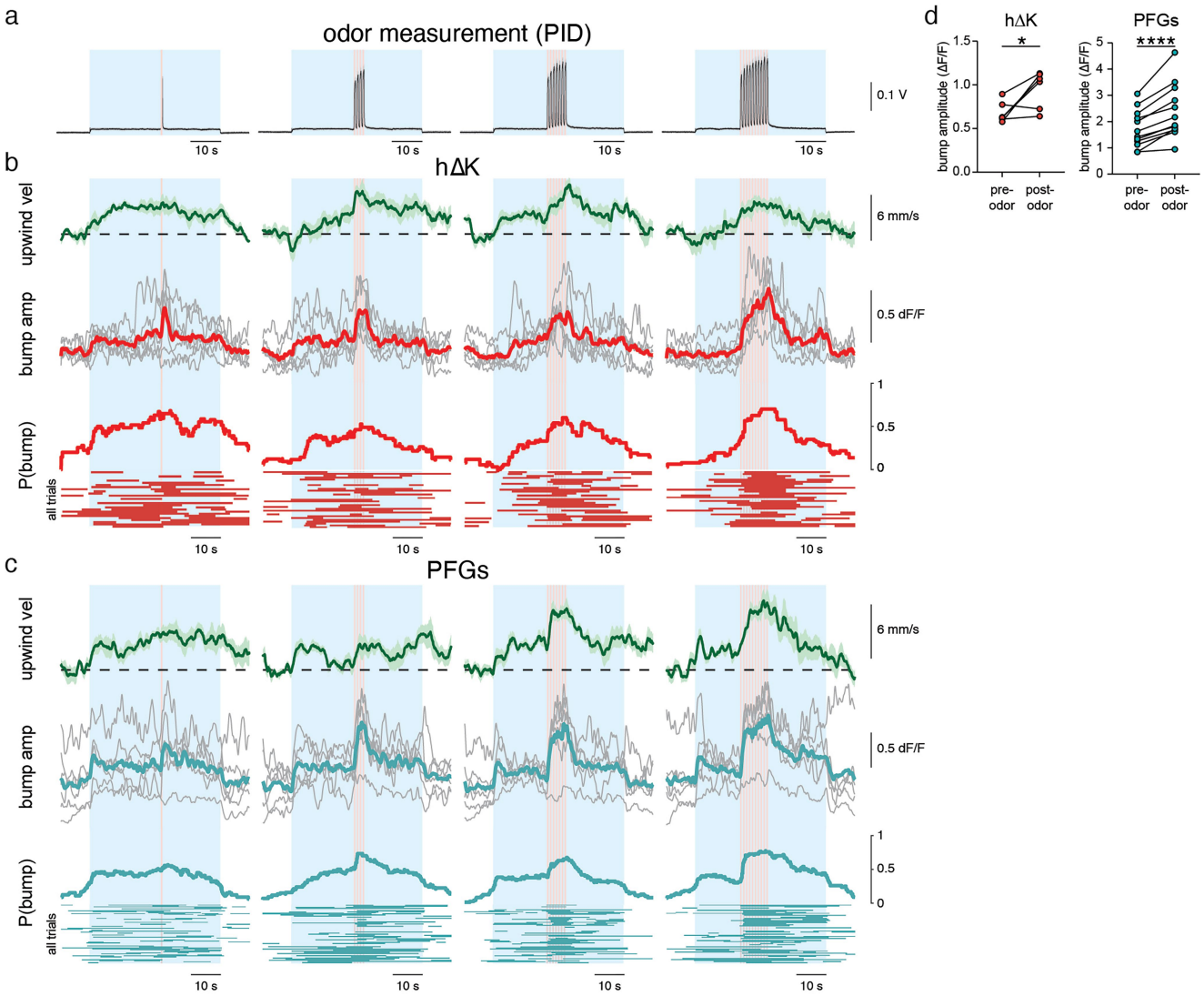
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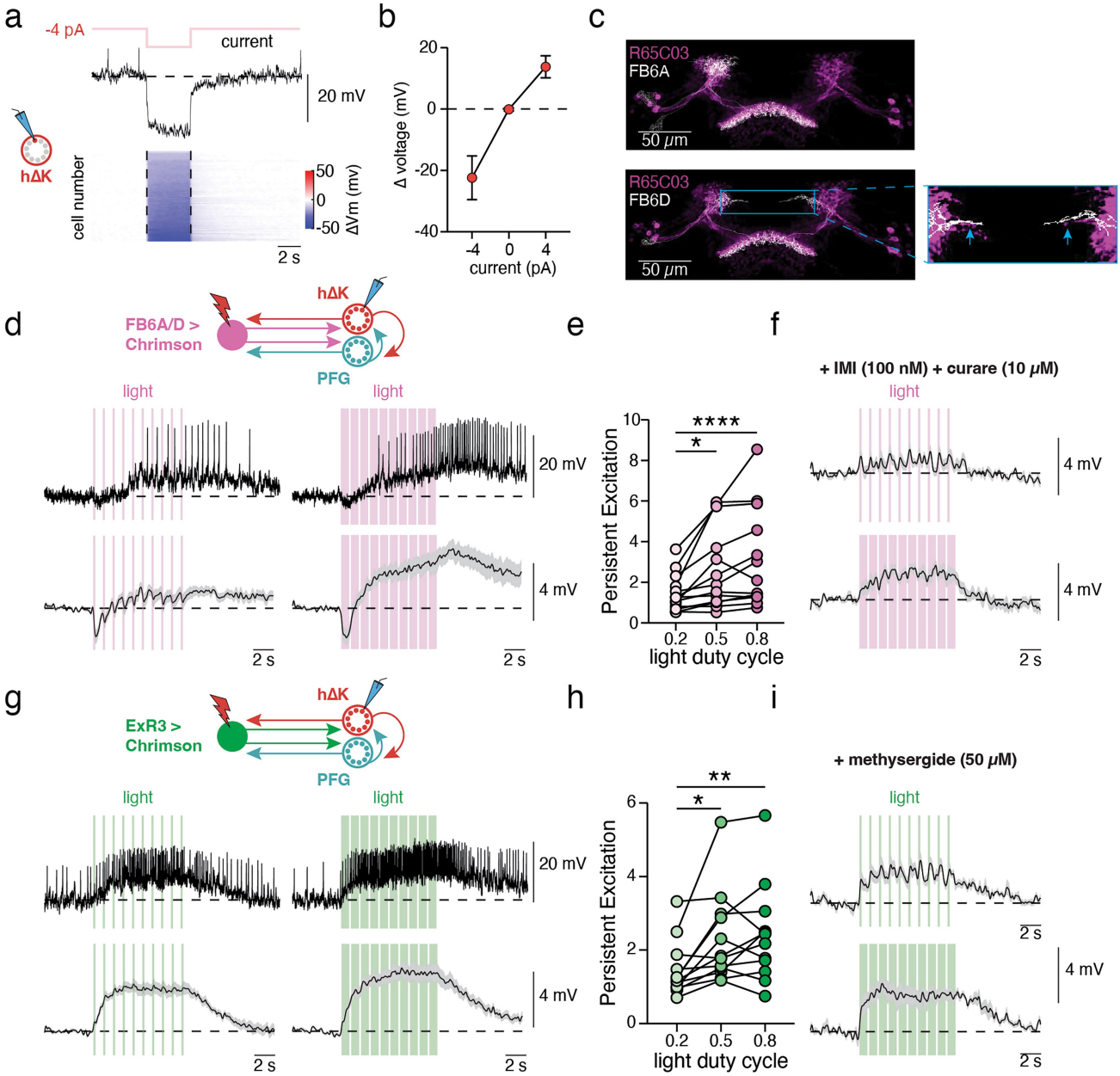
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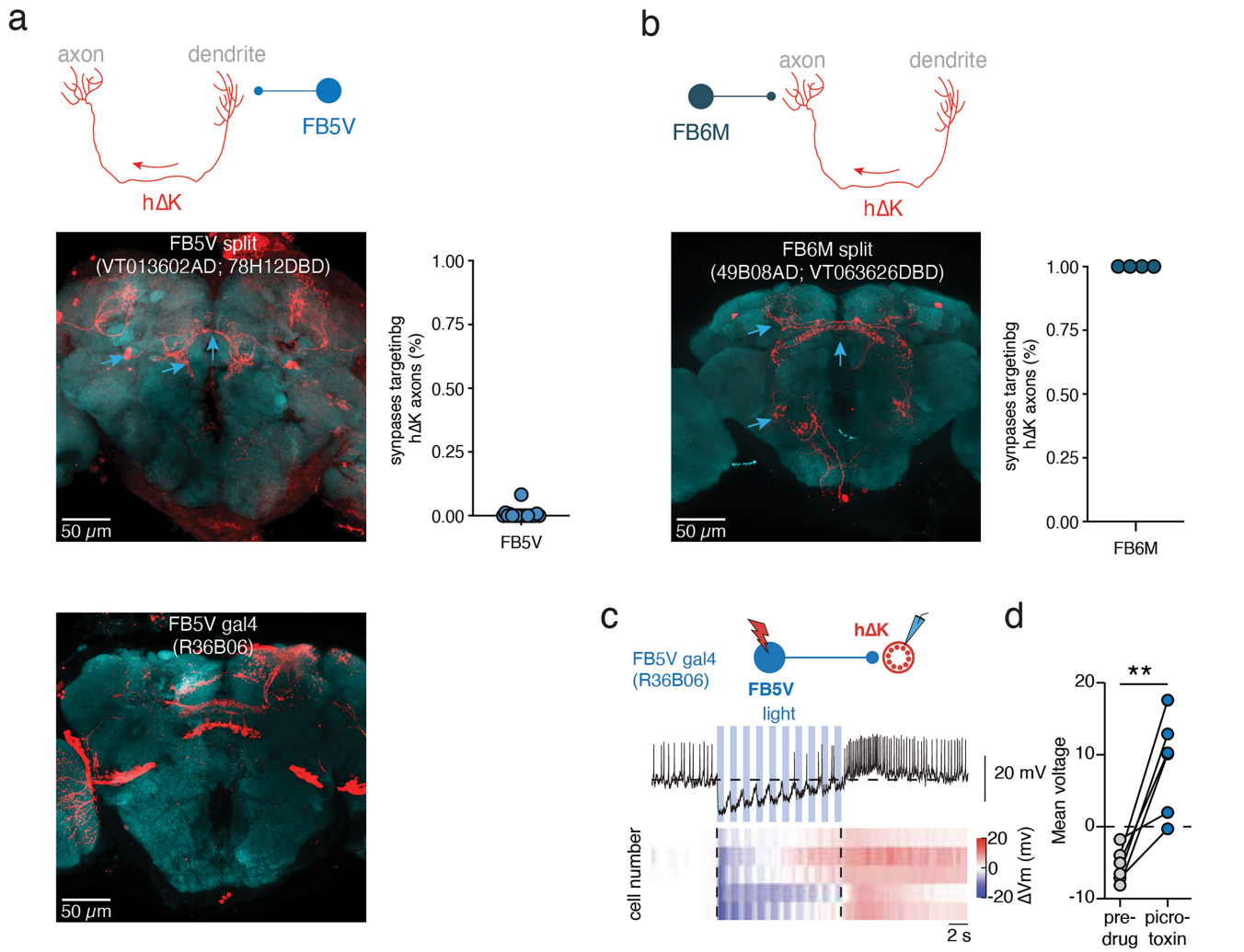
Extended Data Fig. 1 | Additional data on connectomics and neural behavior. a, Morphology of the hΔK population (left) and PFGs population (right) from the FAFB connectome. **b**, Morphology of a connected pair of hΔK and PFG neurons. **c**, Schematic of the axo-axonic connectivity between hΔK (31 neurons) and PFGs (18 neurons). Each hΔK neuron synapse onto fewer PFG neurons than vice versa to account for the difference in population sizes. **d**, Schematic of the connectivity between EPG, PFG, and hΔK neurons. Individual EPG neurons synapse with two sets of PFG neurons in each half of the protocerebral bridge (PB). These PFG neurons reconverge back onto one part of the FB to reciprocally synapse with axons of hΔK neurons. Note, PFG neurons also receive heading inputs from inhibitory Δ7 neurons in the PB, with the same functional phase as EPG inputs (see Fig. 5 for connectivity). **e**, Schematic of the connectivity between FB tangential neurons and hΔK and PFG populations. FB tangential neurons send a single process into the FB that innervates all members of hΔK and/or PFG neurons. **f**, Synaptic weights from PFG to hΔK neurons (left) and hΔK to PFG neurons (right) taken from the hemibrain connectome. Both hΔK and PFG neurons are sorted by their axon position along the FB. **g**, Comparison of the reciprocal synaptic weights between hΔK and PFG neurons. Each dot represents a pair of hΔK and PFG neurons. **h**, Distributions of the total synaptic weight between pairs of recurrently-connected columnar neurons in the central complex. Data are separated for hΔK-PFG pairs (purple) and other columnar pairs (green), which includes all EPG, Δ7, PFN, PFL, PFR, FC, FS, FR, hΔ, and vΔ neurons. **i**, Example virtual trajectories of walking flies before (grey), during (pink), and following (green) odour. Note how flies often maintain their walking direction following odour offset (middle, right). **j**, Distribution of heading before wind/odour (baseline), during odour, 10 s after odour, 20 s after odour, and 30 s after odour (error = mean ± SEM), where 0° indicates upwind. **k**, Left, vectors representing the direction (angle) and duration (length) flies walk following trial start (top, baseline) or odour offset (bottom), with upwards indicating upwind. The black lines represent the mean vector for each condition. Note, in the baseline condition, if flies maintained their direction for long enough following trial start, wind and odour were turned on, leading to the slight bias towards upwind in baseline, particularly for long walks. Right, distributions of the persistence of walking trajectories following trial start (baseline) or odour offset (error = mean ± SEM).



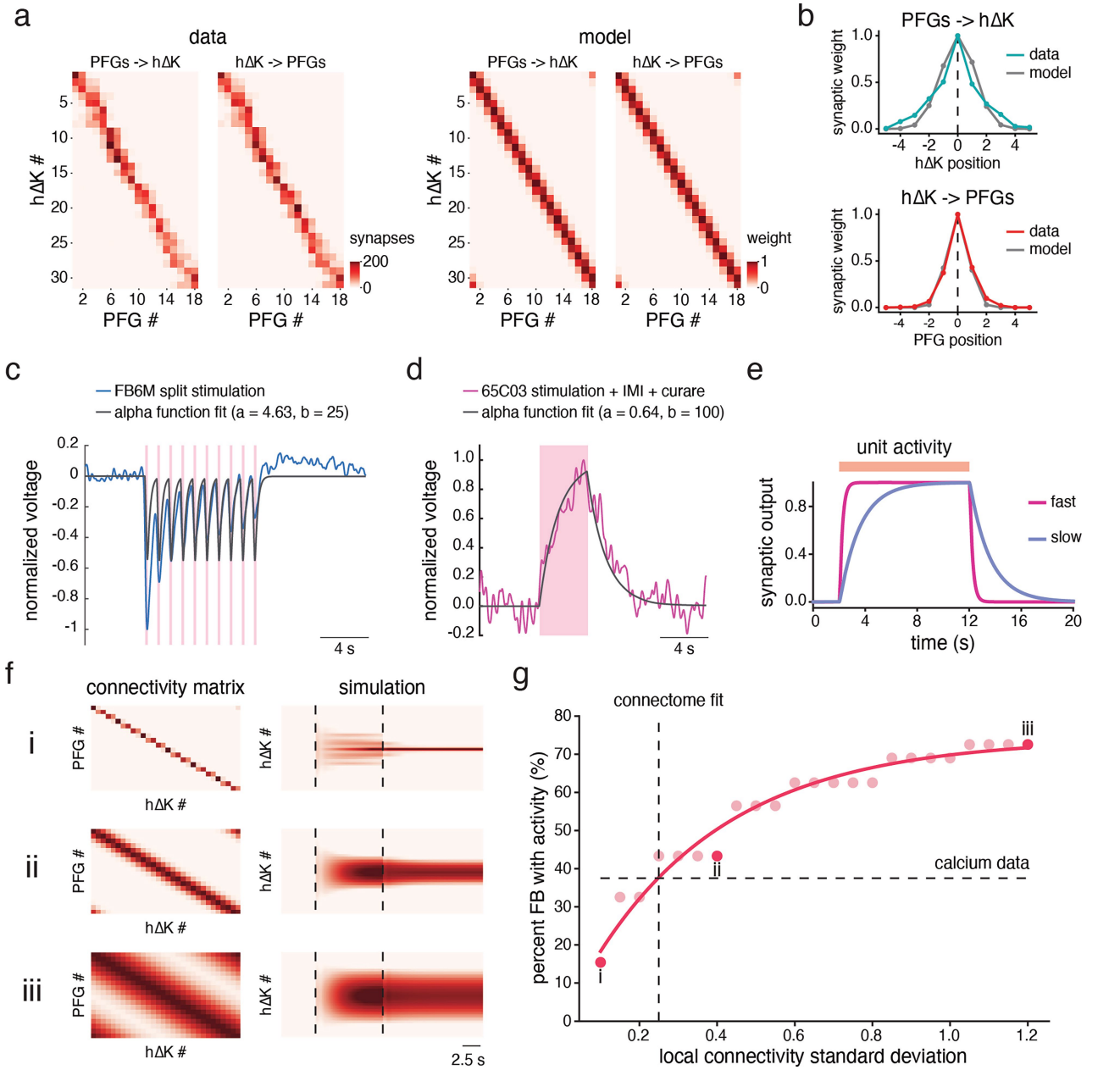
Extended Data Fig. 2 | Additional data on neural responses of hΔK and PFGs. a, Photoionization detector (PID) measurements of odour concentration made at the location of the fly during baseline (no stimuli), wind, and pulses (1, 4, 7, or 10) of odour. **b**, Mean upwind velocity (walking aligned with wind), hΔK bump amplitude, and hΔK bump probability during baseline (no stimuli), wind, and pulses (1, 4, 7, or 10) of odour. Individual fly averages in grey; average across flies in green (behaviour) and red (neural activity). Shaded regions are mean ± SEM. Below P(bump) is a binarized bump data for each trial, where red is bump and transparent is no bump. **c**, Same as **b** but for PFG neurons. Individual fly averages in grey; average across flies in green (behaviour) and blue (neural activity). Below P(bump) is a binarized bump data for each trial, where blue is bump and transparent is no bump. **d**, Average bump amplitude in the 2.5 s before odour onset compared to in the 2.5 s following odour offset for hΔK (left, $n = 6$ flies) and PFG (right, $n = 12$ flies). In all statistical tests: *$0.01 < p < 0.05$, **$0.001 < p < 0.01$, ***$0.0001 < p < 0.001$, ****$p < 0.0001$. Statistical details are in Supplementary Table 1.



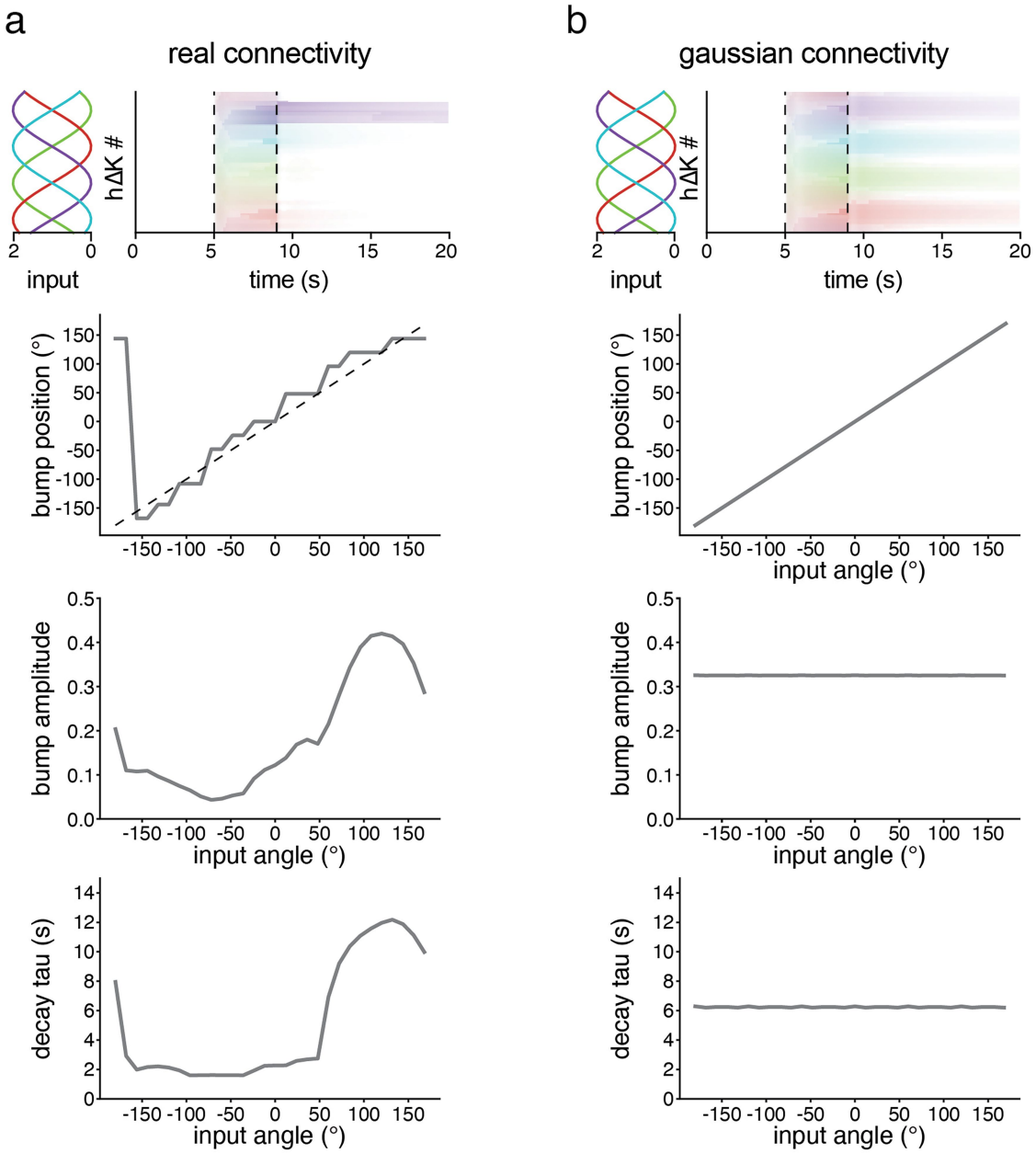
Extended Data Fig. 3 | Additional data on excitatory inputs to hΔK. **a**, Voltage response of hΔK neurons (VT062617-lexA) during a −4 pA current injection. Top: example trace, bottom: heat map of mean membrane potential responses with each cell as a row. **b**, Mean membrane potential responses of hΔK neurons during −4pA, 0pA, or 4pA current injections (n = 67 cells; error = mean ± SEM). **c**, Overlay of R65C03 GAL4 line (image from Flylight[71]) and FB6A (top) or FB6D (bottom) skeletons from the hemibrain connectome, all warped to the Janelia female template brain[72]. The dendrites of FB6A and FB6D overlap strongly with the R65C03 stain. The right shows that R65C03 contains the medial-projecting processes that are unique to FB6D (blue arrows). Note that R65C03 contains multiple tangential neurons (around 17 per side), which overlaps with several other layer 6 neurons, such as FB6C_a, FB6C_b, FB6E, FB6G, FB6I, and FB6Z. **d**, Spiking and membrane potential responses of hΔK neurons to optogenetic activation of FB6A/D (65C03-gal4) using pulsed light (1 Hz) at 0.2 (left) or 0.8 (right) duty cycle (error = mean ± SEM). **e**, Mean persistent excitation (see Methods) following activation of FB6A/D at 0.2, 0.5, or 0.8 duty cycle (n = 13 cells). Data for 0.5 duty cycle are from Fig. 2c. **f**, Mean membrane potential responses of hΔK neurons in response to optogenetic activation of FB6A/D using pulsed light (1 Hz) at 0.2 (top) or 0.8 (bottom) duty cycle in the presence of IMI (100 nM) and curare (10 μM) (error = mean ± SEM). **g**, Same as **d** but in response to activating ExR3 neurons (24C07-gal4) (error = mean ± SEM). **h**, Same as **e**, but in response to activating ExR3 neurons (n = 13 cells). Data for 0.5 duty cycle are from Fig. 2g. **i**, Mean membrane potential of hΔK neurons in response to optogenetic activation of ExR3 using pulsed light (1 Hz) at 0.2 (top) or 0.8 (bottom) duty cycle in the presence of the general serotonin receptor antagonist methysergide (50 μM) (error = mean ± SEM). In all statistical tests: *0.01 < p < 0.05, **0.001 < p < 0.01, ***0.0001 < p < 0.001, ****p < 0.0001. Statistical details are in Supplementary Table 1.



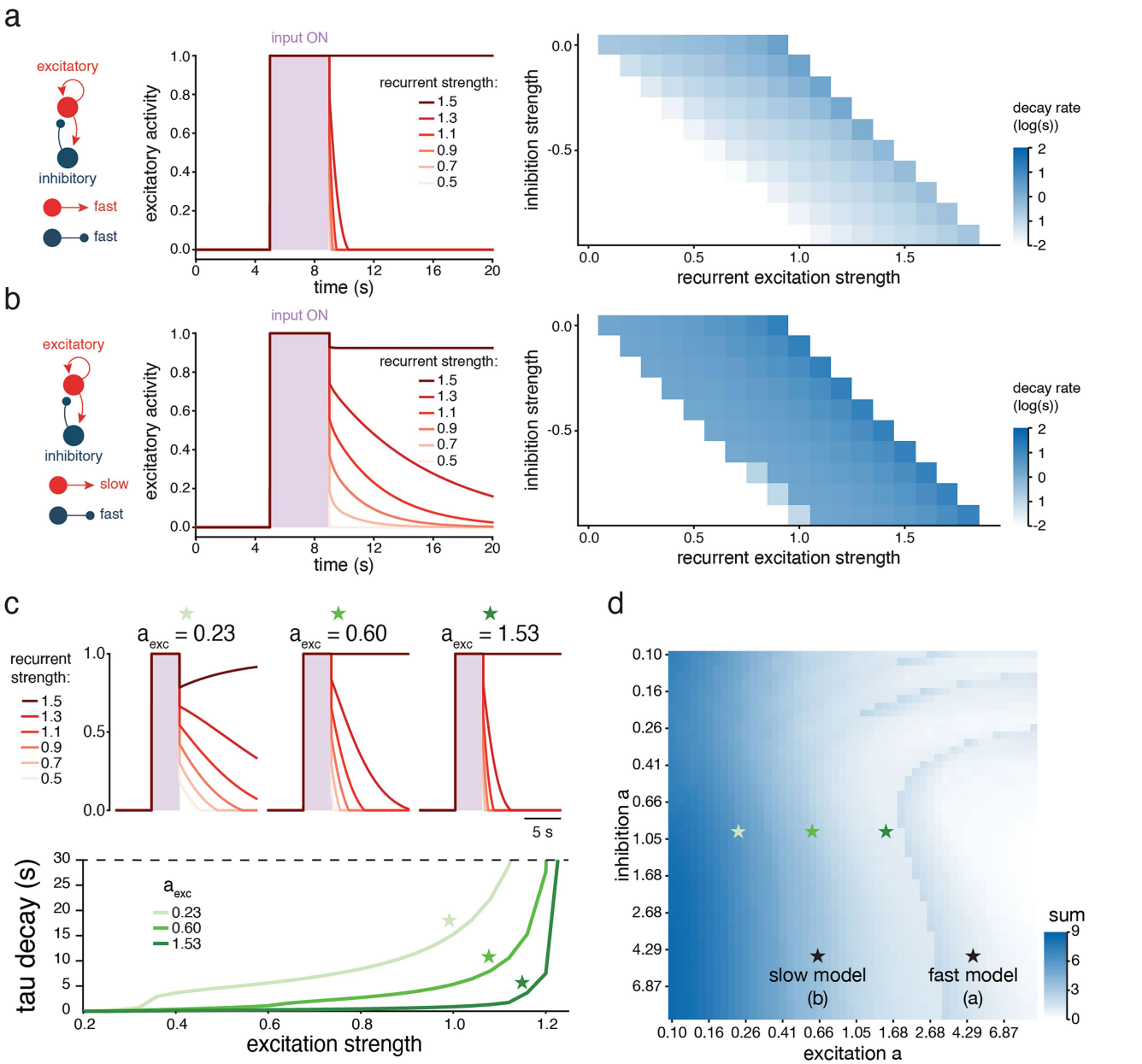
Extended Data Fig. 4 | Additional data on inhibitory inputs to hΔK. **a**, Top, schematic showing axo-dendritic connectivity between FB5V and hΔK neurons. Middle, left, GFP staining of the line VT013602AD; 78H12DBD, which targets FB5V, among other neurons. Middle, right, percent of FB5V synapses targeting the hΔK axons. Each data point corresponds with one FB5V neuron. Bottom, GFP staining of the line R36B06-gal4, which targets FB5V, among other neurons. **b**, Top, schematic showing axo-axonic connectivity between FB6M and hΔK neurons. Bottom, left, GFP staining of the line 49B08AD; VT063626DBD, which targets FB6M. Arrows indicate characteristic morphological features of FB6M, including FB axons, ventral WED dendrites, and dorsal SMP dendrites. Bottom, right, percent of FB6M synapses targeting hΔK axons. Each data point corresponds with one FB6M neuron. **c**, Spiking and membrane potential response of hΔK neurons to optogenetic activation of FB5V (36B06-Gal4). Top: example trace; bottom: heat map of the mean membrane potential response with each cell as a row. **d**, Mean voltage responses of hΔK neurons in response to optogenetic activation of FB5V (36B06-Gal4) before and after the application of picrotoxin (5 μM) ($n = 6$ cells). In all statistical tests: $^{*}0.01 < p < 0.05$, $^{**}0.001 < p < 0.01$, $^{***}0.0001 < p < 0.001$, $^{****}p < 0.0001$. Statistical details are in Supplementary Table 1.



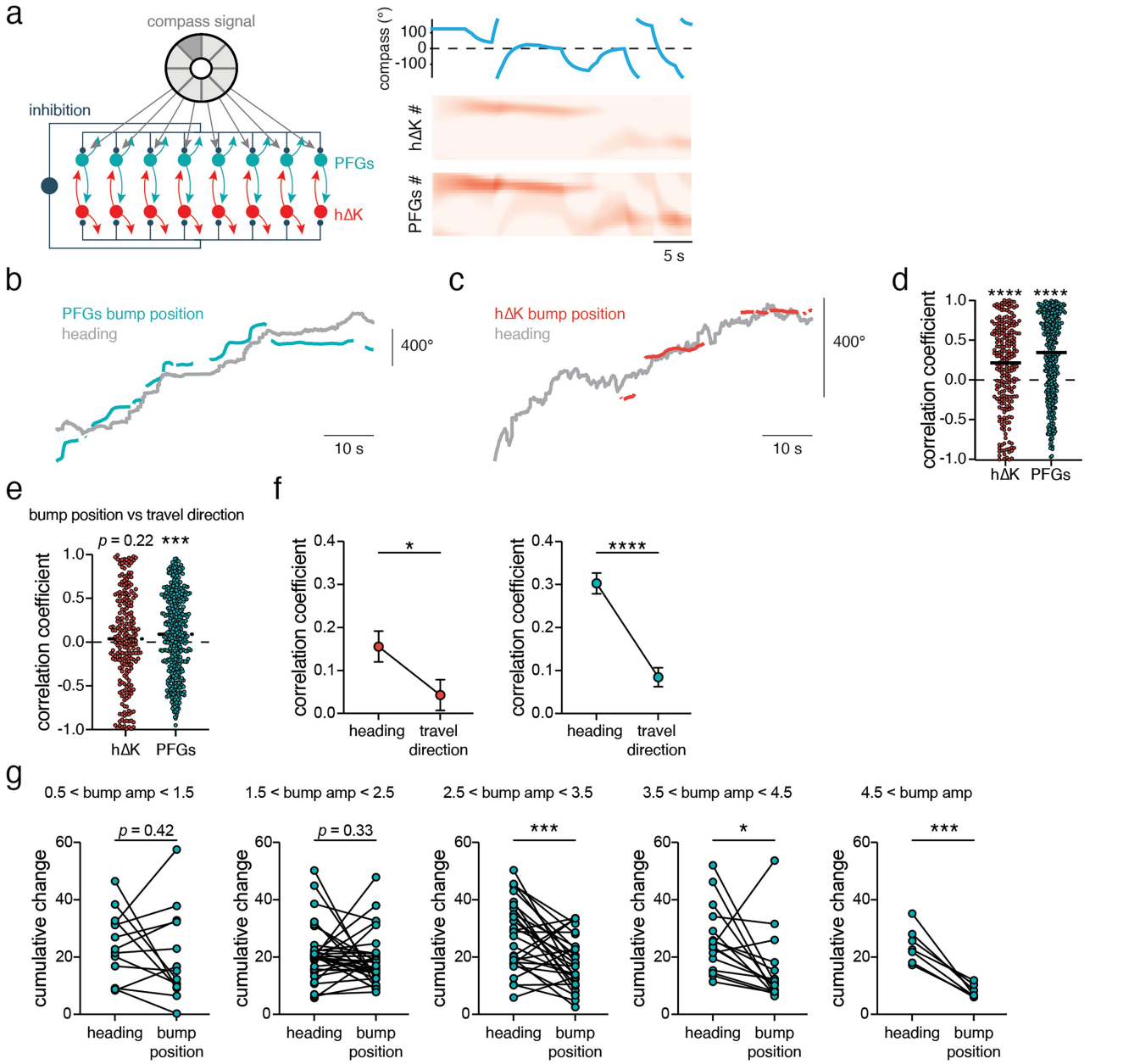
Extended Data Fig. 5 | Methods details for recurrent model of hΔK and PFGs. a, Comparison of the real connectivity between hΔK and PFG neurons (left) and the Gaussian connectivity in the best fit model (right, σ = 0.25 radians). **b**, Local connectivity between hΔK and PFG neurons in the data and model, visualized by aligning each presynaptic neuron and averaging the synaptic weights to neighbouring postsynaptic partners. **c**, Alpha function fit to hΔK voltage responses to pulsed (0.2 duty cycle) stimulation of FB6M (49B08AD; VT063626DBD). **d**, Alpha function fit to hΔK voltage responses to a single 4 s stimulation of FB6A/D (65C03-gal4) in the presence of the nicotinic acetylcholine receptor blockers curare (10 μM) and IMI (100 nM). These blockers are included to remove acetylcholine signaling by FB6A/D and hΔK neurons, leaving only the slow synaptic signalling from FB6A/D for fitting. **e**, Synaptic output of model neurons using fast (pink) or slow (blue) synaptic speeds in response to a prolonged step of activity. **f**, Simulations of the hΔK-PFG network performed across various local connectivity widths. The Gaussian connectivity between hΔK and PFG neurons is shown on the left and the simulation results are shown on the right. Note how the bump width increases as the standard deviation of connectivity is increased. **g**, Width of attractor bumps (percent of the FB with activity) as a function of the standard deviation of recurrent connectivity (see Methods). Red line is an exponential plateau function fit to the simulation data ($y = a*(1 - e^{-bx})$, a = 74, b = 2.83). The percent of FB with calcium activity during hΔK bumps and the local connectivity of the best fit Gaussian model are shown as dashed lines.



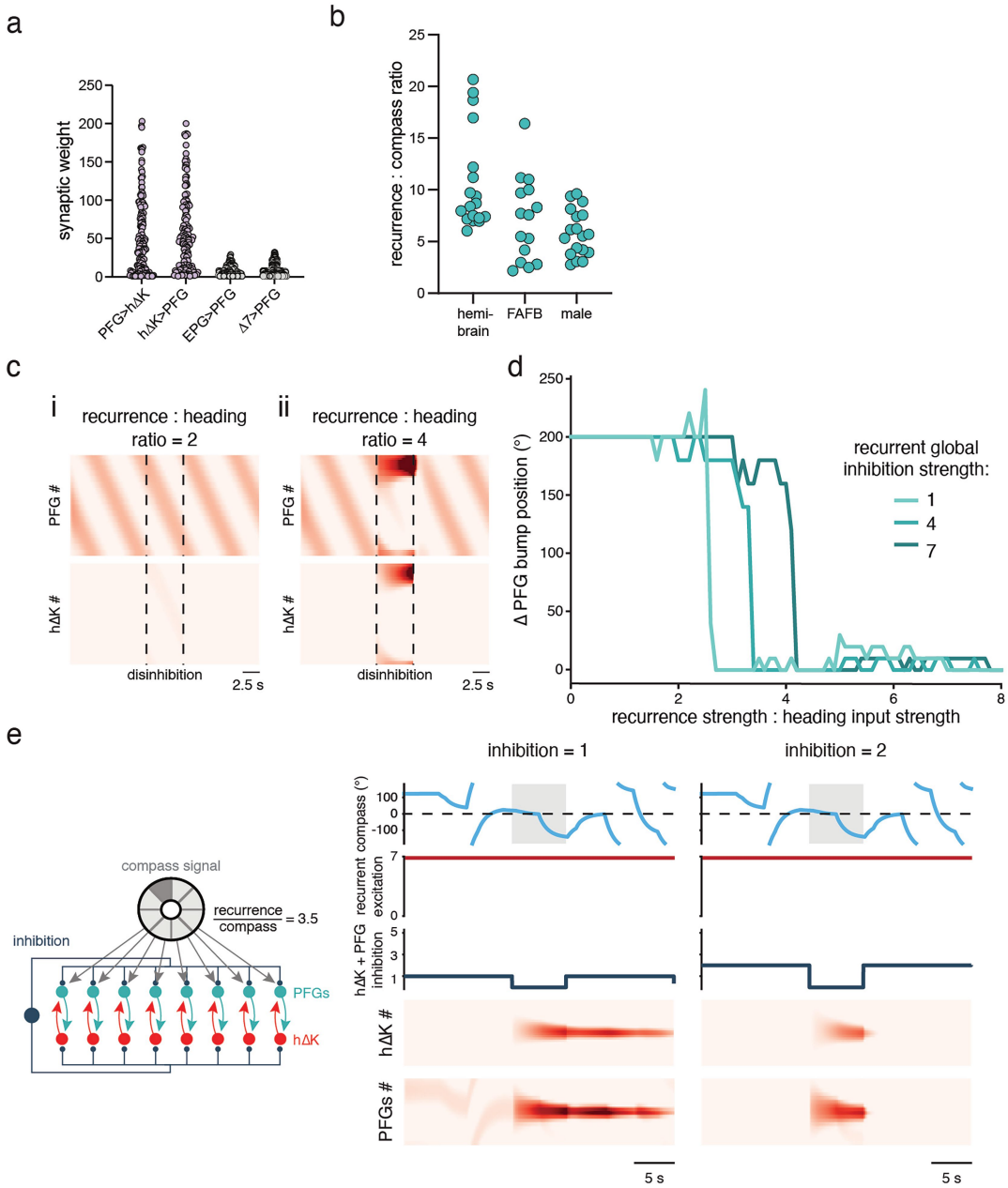
Extended Data Fig. 6 | A model with real connectivity between PFG and hΔK can form a bump at most locations but shows inhomogeneity in bump amplitude and duration. a, hΔK bump position, bump amplitude, and decay rate (tau) in response to sinusoidal input across all possible phases when the connectivity between hΔK and PFG neurons is taken from the hemibrain connectome (see Extended Data Fig. 1f). Top heatmap shows four separate simulations, all overlayed, in response to the sinusoidal inputs shown. Note that bump position can be stabilized at most positions, but the bump amplitude and decay time constant depend strongly on position. **b**, Same as **a** but using connections approximated by a Gaussian model.



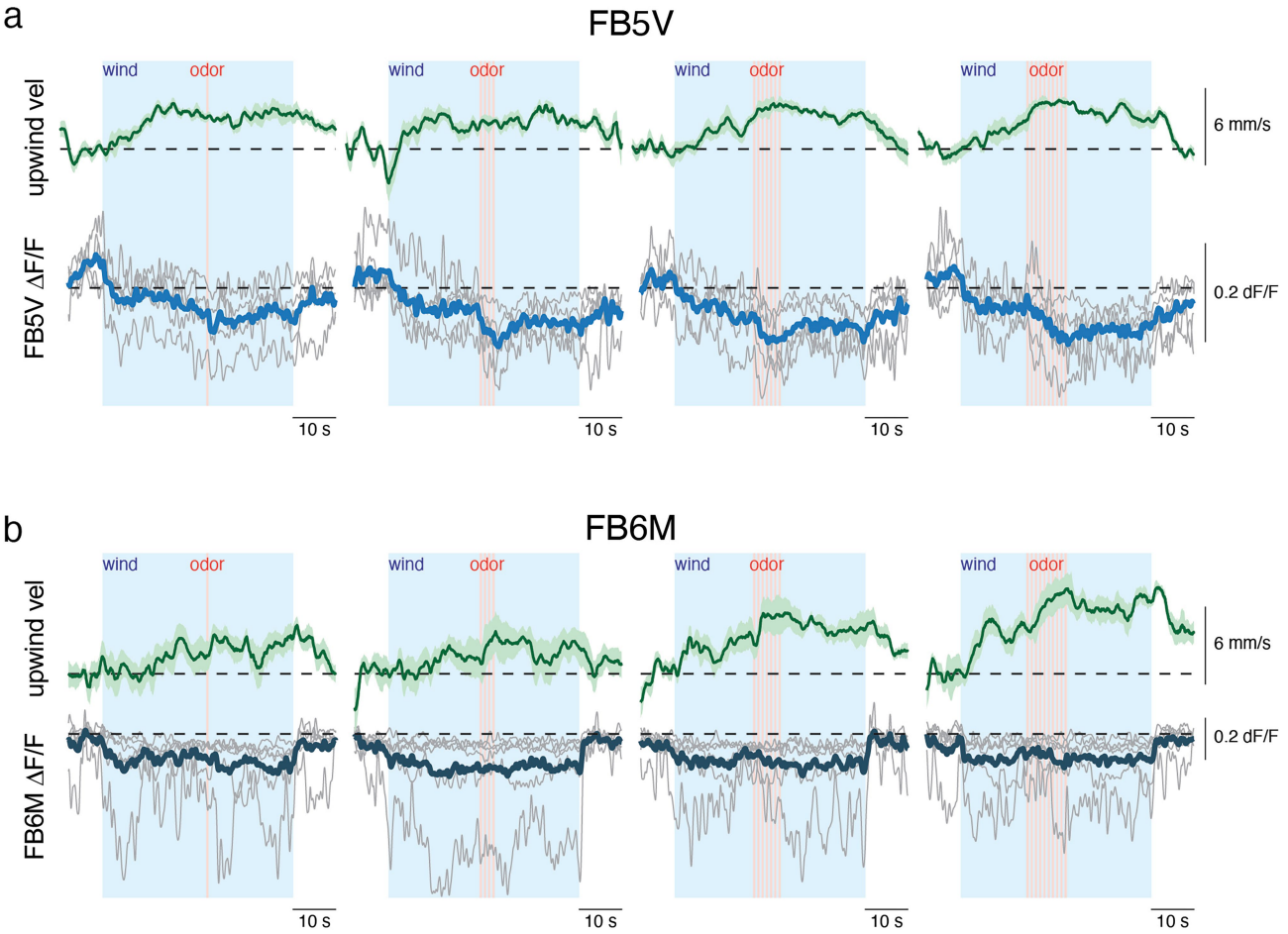
Extended Data Fig. 7 | Slow excitation stabilizes persistent activity in a minimal recurrent network. a, Left, schematic of a simple two-unit recurrent network using fast excitation and fast excitation. Middle, activity of excitatory unit in response to inputs as a function of recurrent excitation strength. Right, parameter space showing the decay rate (log_{10} scale) of the excitatory unit following input offset as a function of recurrent excitation and inhibition strength. **b**, Same as **a** but the excitatory unit uses slow synaptic signaling. **c**, Two-unit model simulations as the excitatory synaptic decay rate (a) is slowed (a = 0.23, 0.60, or 1.53) and the recurrent excitatory strength is increased. Top, simulation output of the excitatory unit across conditions. Bottom, excitatory decay tau for each synaptic speed across recurrent excitation strengths. As the excitatory synaptic decay rate slows, the transition between fast activity decay and zero activity decay occurs more gradually. **d**, Parameter space showing the sum of the excitatory tau decay across excitation strengths (sum of each curve in **c**) across various speeds of excitation and inhibition. The models used in **a, b**, and **c** are all shown as stars.



Extended Data Fig. 8 | Additional data on differences in bump dynamics between PFG and hΔK. a, Simulations of hΔK and PFG population activity when the EPG compass signal (light blue) is randomly and stochastically updated and PFG neurons receive direct sinusoidal inputs from the compass. Left shows a schematic of the network configuration for the simulation. Right shows the simulation output. Note presence of a bump at all times in both populations, and occasionally the presence of multiple coexisting bumps in PFG neurons. **b**, Example of the PFG bump position aligned to the corresponding heading of the fly. **c**, Same as **b** but for hΔK neurons. **d**, Correlation coefficients between heading and hΔK (red, $n = 231$ trials) or PFG (blue, $n = 447$ trials) bump position across trials. **e**, Correlation coefficients between travel direction and hΔK (red, $n = 231$ trials) or PFG (blue, $n = 447$ trials) bump positions across trials. **f**, Average correlation coefficients of hΔK (left, $n = 231$ trials) or PFG (right, $n = 447$ trials) bump position with heading direction versus travel direction (error = mean ± SEM). Correlation compared to 0 using a two-sided Wilcoxon signed rank test. **g**, Cumulative change in heading and PFG bump position during wind shift trials binned according to the mean PFG bump amplitude. Each pair of dots represents the heading shift and bump position shift for a single wind shift trial ($n_{0.5<amp<1.5} = 13$ trials, $n_{1.5<amp<2.5} = 33$ trials, $n_{2.5<amp<3.5} = 29$ trials, $n_{3.5<amp<4.5} = 16$ trials, $n_{4.5<amp} = 7$ trials). Cumulative change compared between heading shifts and bump position shifts using two-sided paired t-tests. In all statistical tests: $^*0.01<p<0.05$, $^{**}0.001<p<0.01$, $^{***}0.0001<p<0.001$, $^{****}p<0.0001$. Statistical details are in Supplementary Table 1.



Extended Data Fig. 9 | Additional data on disinhibition model. a, Synaptic weights between pairs of PFG and hΔK neurons, as well as from EPG and Δ7 neurons to PFG neurons from the hemibrain connectome. **b**, The ratio of the average synaptic input between hΔK neurons and PFG neurons (recurrent) to total synaptic input from EPGs (compass), for each PFG neuron. The average (mean ± SD) ratios for each connectome are 10.70 ± 4.816 (hemibrain, $n = 18$), 7.15 ± 4.06 (FAFB, $n = 15$), and 5.84 ± 2.23 (male CNS, $n = 18$). **c**, Simulation of hΔK and PFG population activity with a disinhibitory gate of hΔK neurons when the recurrence: heading ratio is low (ratio = 2) or high (ratio = 7). The global recurrent inhibition strength is 1 for both simulations. Note the bump stabilization only occurs at the higher ratio. **d**, Change in PFG bump position during disinhibition as the recurrence: heading ratio increases, and as the recurrent global inhibition strength changes. Note the same heading signal is used during each simulation, which is a constant rotation, as seen in **c**. **e**, Simulation of hΔK and PFG population activity with inhibition to both PFG and hΔK. As in Fig. 5, EPG compass signal (light blue) is randomly and stochastically updated and global inhibition to hΔK and PFG neurons is transiently shut off. The global inhibition strength was set to 1 (left) or 2 (right), showing how the simulations gate activity in both hΔK and PFG neurons but do not permit continuous compass representations in PFG neurons during inhibition.



Extended Data Fig. 10 | Additional data on encoding of sensory input by FB5V and FB6M. a, Mean upwind velocity and FB5V ΔF/F during baseline (no stimuli), wind, and pulses (1, 4, 7, or 10) of odour. Individual fly averages in grey; average across flies in green (behaviour) and blue (neural activity). Shaded regions are mean ± SEM. **b**, Same as **a** but for FB6M neurons. Individual fly averages in grey; average across flies in green (behaviour) and blue (neural activity).

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☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	All custom Python and Matlab code for data collection is available on Github at https://github.com/nagellab/Lanzetal2025 Fly trajectories on the walking ball were tracked using Fictrac version 2.00
Data analysis	All custom Python and Matlab code for data analysis is available on Github at https://github.com/nagellab/Lanzetal2025 Data analysis was performed in Matlab R2024b Connectomic analysis and computational modeling were performed using Python 3.7.7

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

All data collected during this study are available at 10.5281/zenodo.19687299

In addition, the following databases were used in this study:
- Hemibrain connectome, hemibrain:v1.2.1, http://neuprint.janelia.org
- FAFB connectome, FAFB v783, https://codex.flywire.ai
- Male CNS connectome, male-cns:v0.9m, http://neuprint.janelia.org

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	n/a
Reporting on race, ethnicity, or other socially relevant groupings	n/a
Population characteristics	n/a
Recruitment	n/a
Ethics oversight	n/a

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	Sample sizes for imaging and behavioral experiments were based on Kathman et al. 2026. Sample size electrophysiology were based on Suver et al. 2019 and Currier et al. 2020.
Data exclusions	For imaging experiments, flies were only included for further analysis if they reliably oriented upwind on at least 50% of trials, which was identified by a convergence of heading values towards the upwind direction (0°) and an increase in mean upwind velocity around wind or odor onset. For electrophysiology experiments, cells were only included for analysis if they showed visible spiking activity, defined as rapid deflections above 10 mV. Other than these criteria, all data and replications were included in the manuscript.
Replication	Imaging experiments in behaving flies were performed in at least 6 flies per genotype. Electrophysiology experiments were performed 5-15 cells (generally one per fly) for each genotype.
Randomization	All stimuli for behavior and imaging were presented in random order within a block of trials. All comparisons were based on genotype or pharmacological treatment and all flies of a given genotype or treatment were included in each sample.
Blinding	All analyses for behavior and imaging data were automated and applied equally to all genotypes make genotype blinding unnecessary. Blinding was often impossible because genotype could be determined visually from marker phenotypes indicating the presence or absence of balancers, or cell type could be determined visually during imaging or electrophysiology. The same code was applied to all genotypes to produce the analysis, therefore the experimenter's knowledge of the genotype was irrelevant to the outcome of the experiment.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☐	☒ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

chicken anti-GFP Fisher Scientific RRID:AB_1074893
mouse anti-nc82 DSHB RRID:AB_2314866
rabbit anti-DsRed Clontech 632496
Alexa488-conjugated goat anti-chicken Fisher Scientific RRID:AB_2534096
Alexa633-conjugated goat anti-mouse Fisher Scientific RRID:AB_2535719
Alexa568-conjugated goat anti-rabbit Fisher Scientific RRID:AB_2315774
Alexa568-conjugated streptavidin Fisher Scientific RRID:AB_2576217

Validation

Antibodies were validated by staining driver lines with known expression. Antibodies were used only to identify the anatomy of labeled neurons and not to identify molecular species or localization. GFP antibody has been used in WB, IHC, and ICC/IF in at least 17 confirmed publications according to ThermoFisher Scientific: https://www.thermofisher.com/antibody/product/GFP-Antibody-Polyclonal/PA1-9533. nc82 antibody has been cited 1201 times according to DSHB: https://dshb.biology.uiowa.edu/nc82. ds-Red antibody has been cited 2618 times according to Takara Bio (previously Clontech): https://www.takarabio.com/products/antibodies-and-elisa/fluorescent-protein-antibodies/red-fluorescent-protein-antibodies?srsltid=AfmBOop1hxSp7GD1OPh1oBI0ynhI9cXuk1yLlAfzN4G1gnFE95m-pwWU.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals	Drosophila melanogaster, aged 1-14 days. Genotypes described in detail in Methods
Wild animals	n/a
Reporting on sex	All experiments were performed in female flies.
Field-collected samples	n/a
Ethics oversight	Only invertebrate organisms were used in this study, therefore no animal protocol was required.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Plants

Seed stocks	n/a
Novel plant genotypes	n/a
Authentication	n/a